



REVIEW ARTICLE

Farmed yellow mealworm (*Tenebrio molitor*; Coleoptera: Tenebrionidae) welfare: species-specific recommendations for a global industry

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Abstract

Yellow mealworms (*Tenebrio molitor*; Coleoptera: Tenebrionidae) are currently the most farmed holometabolous insect species in the insects as food and feed industry, with over 300 billion individual mealworms reared annually. Yellow mealworm larvae are being developed for potential uses as human protein, pet, livestock and fish feed, reclamation of mycotoxin-contaminated grains, and more. Insect welfare is of great interest to consumers, producers, and academics; yet no studies have considered the species-specific welfare concerns of farmed yellow mealworms under current industry conditions. Following a model for considering farmed insect welfare, we review yellow mealworm biology and its relationships to welfare in commercial rearing facilities, including: interspecific interactions (predators, parasites, and pathogens), abiotic conditions (temperature, hydration, atmospheric gasses, lighting), nutrition (including pollutants, plastics, and hormones), intraspecific concerns (genetics, morphological defects, cannibalism, density, mating and oviposition needs, and handling-associated stress), and slaughter and depopulation methods (including anesthesia and stunning). From this review, we identify practical recommendations for improving current welfare concerns in the industry and mitigating future concerns that may appear as the industry continues to grow. Finally, we discuss future research directions that are necessary to better understand the welfare of this species.

Keywords

insect welfare – rearing conditions – survival – slaughter methods – mealworm

1 Introduction

The insects as food and feed (IAFF) industry aims to use insects to solve global protein and sustainability challenges. Insects are high in protein, have a smaller environmental footprint than vertebrate livestock, can contribute to local, circular food systems by turning waste

products into proteins, oils, fuel, and more, and may also promote economic growth and food security in developing nations (Chia *et al.*, 2019; de Jong and Nikolik, 2021; Grau *et al.*, 2017; Lee *et al.*, 2022; Oonincx and de Boer, 2012; Rumbos *et al.*, 2022; van Huis and Tomberlin, 2017; van Huis and Gasco, 2023).

As of 2020, the IAFF industry farmed over 1 trillion individual insects a year, producing approximately 60–70,000 tons of insect protein (Rowe, 2020). Mass-rearing facilities, currently a small share of global production, are responsible for ~10,000 metric tons. By 2030, these facilities may produce up to 500,000 metric tons of insect protein (de Jong and Nikolik, 2021), producing >8 trillion individuals annually. Aquaculture is expected to produce the largest increase in demand for farmed insects (de Jong and Nikolik, 2021; Makkar *et al.*, 2014). Even if insects replace only 25% of fishmeal, 4–5 million metric tons (~60–80 trillion individual insects) would need to be farmed each year (de Jong and Nikolik, 2021; Rowe, 2020). The growth of the IAFF industry represents one of the largest undertakings of industrial livestock rearing in history.

The IAFF industry is interested in insect welfare for a variety of economic, as well as potentially ethical, reasons. Considering livestock welfare can contribute to improving production outcomes, provide opportunities for product differentiation, and help the industry maintain a social license to operate in the face of growing public interest in insect pain and welfare (reviewed in Barrett and Adcock, 2023). Although there is considerable empirical debate on whether insects are sentient, there appears to be increasing evidence for insect pain (reviewed in Gibbons *et al.*, 2022; and see Lambert *et al.*, 2021), suggesting that insect producers are also considering precautionary, ethical reasons to protect the welfare of their animals. All these concerns have motivated urgent calls for species-specific investigations of insect welfare by producers, consumers, and industry-associated academics (Bear, 2019, 2021; Boppré and Vane-Wright, 2019; de Goede *et al.*, 2013; Delvendahl *et al.*, 2022; Erens *et al.*, 2012; Fukuda *et al.*, 2023; Gjerris *et al.*, 2016; IPIFF, 2019; van Huis, 2021).

So far, the only species-specific investigations of farmed insect welfare have involved the Diptera (black soldier flies: Barrett *et al.* 2022; black soldier flies and house flies: Kortsmits *et al.*, 2022; Voulgari-Kokota *et al.*, 2023) and Hymenoptera (honey bees: Garrido and Nanetti, 2019). However at least 290 billion yellow mealworms (*Tenebrio molitor*; Coleoptera: Tenebrionidae; hereafter YM) are farmed annually (Rowe, 2020). They are the most frequently mentioned ‘edible insect’ in the published literature (Rumbos and Athanassiou 2021). The larvae (hereafter, YML) may be reared for human consumption (though concerns with allergenicity and safety must still be considered; Barre *et al.*, 2019; Bordiean *et al.*, 2020; Cappelli *et al.*, 2020; Garino *et al.*, 2020; van Huis and Tomberlin, 2017) and also reared for

chicken and pig diets (Hong *et al.* 2020), or to replace fishmeal (reviewed in Shafique *et al.*, 2021). If concerns associated with cost, scale, benzoquinone toxicity, and contaminants can be addressed (de Jong and Nikolik, 2021; Grau *et al.*, 2017; Shafique *et al.*, 2021), YML are likely to play a significant role in the fishmeal replacement that is driving industry growth. This necessitates a review of YM welfare in farmed conditions.

Barrett *et al.* (2022) introduced a model for reviewing species-specific insect welfare concerns. Here, we apply this approach to YM, generating the first comprehensive, species-specific paper on YM welfare in industry conditions.

2 Welfare framework

An animal’s welfare is a characterization of how well that animal is doing relative to its own interests. Welfare frameworks use empirical criteria to assess animal welfare. The UK’s Farmed Animal Welfare Council proposed the now-standard version of the “Five Freedoms” framework, which operationalizes animal welfare as: freedom from thirst, hunger, and malnutrition; freedom from discomfort; freedom from pain, injury and disease; freedom from fear and distress; and freedom to express normal behavior (Brambell, 1965). This framework has been widely adopted, has the force of law in some jurisdictions (e.g. in much of Europe due to the EU’s Animal Act; see Council Directive 98/58/EC), and has already been applied to insects (Barrett *et al.*, 2022; Erens *et al.*, 2012, etc.).

However, there are some limitations to the Five Freedoms framework. Perhaps the most discussed is that the Five Freedoms framework focuses largely on negative dimensions of welfare (again, hunger, discomfort, pain), leaving little room for the various dimensions of positive welfare (Green and Mellor, 2011).

For this reason, many animal welfare scientists have shifted to the closely-related “Five Domains” framework (Figure 1), which operationalizes welfare as four physical/functional domains that impact the one mental domain, and where all five domains have both positive and negative dimensions (Mellor and Reid, 1994).

More importantly for present purposes, though, the Five Domains framework is especially useful for assessing the welfare of relatively understudied species, such as insects. In light of the many unknowns about insects’ welfare, a welfare assessment framework can advance preliminary inquiries into their welfare by suggesting classes of proxies that have proven useful in better-

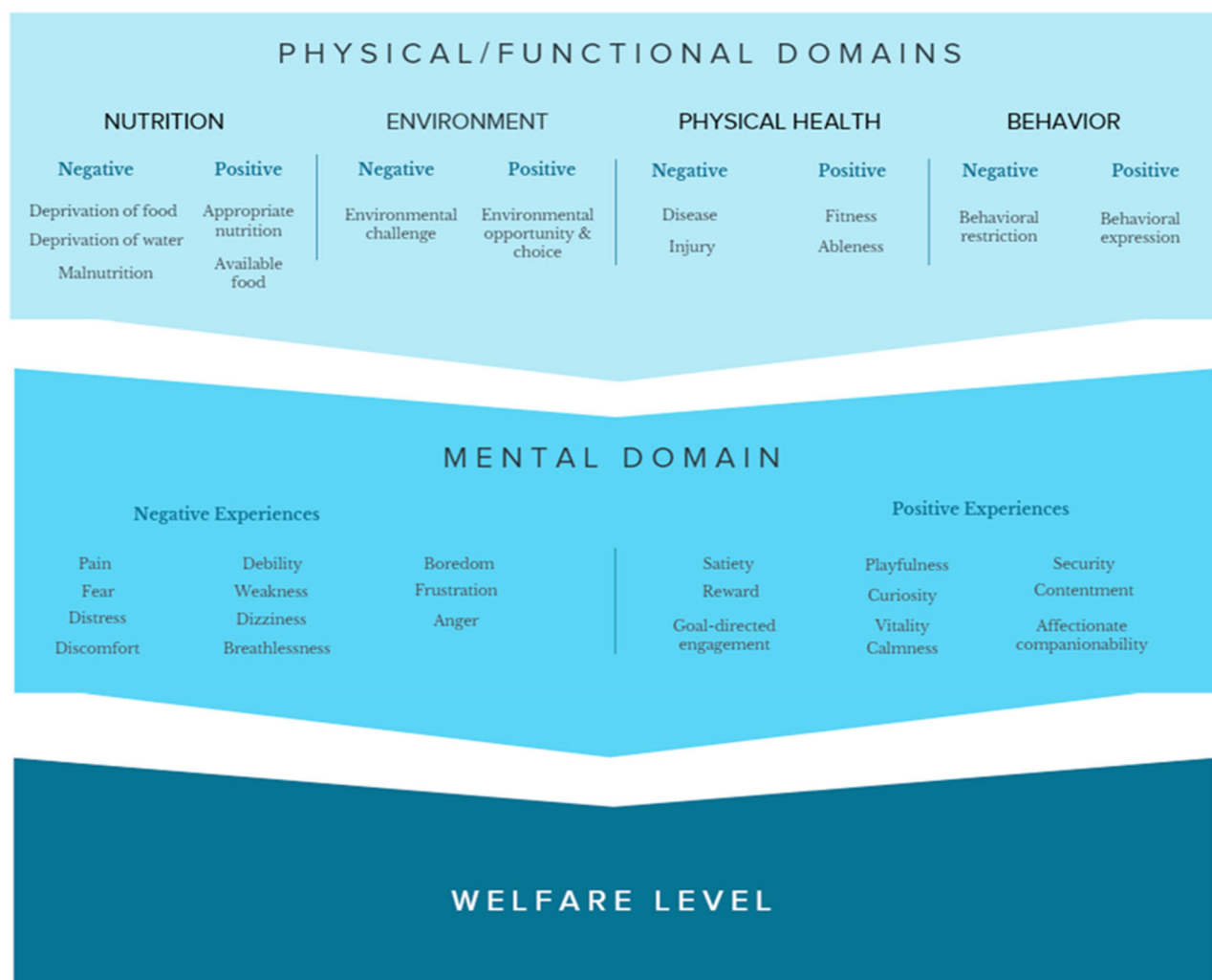


FIGURE 1 The Five Domains framework of animal welfare.

understood species. That is, by directing attention toward nutrition, environment, physical health, and behavior, the Five Domains framework helps guide tentative welfare assessments in the face of considerable uncertainties about what affects YM's interests.

Accordingly, we have organized our review around these domains (while recognizing that many welfare issues could be framed in terms of more than one of them). Where appropriate, we make some cautious suggestions about how the mental domain may be affected by positive and negative dimensions of the four physical/functional domains. However, because of current uncertainties regarding the specific relationships between dimensions of each physical/functional domain and insects' mental states, we focus on the former, as those are the domains about which it is possible to make claims with a higher degree of confidence.

Adapted from Mellor *et al.* (2015); originally published in Barrett and Fischer (2023)

As the Five Domains framework was developed for vertebrates, it still must be applied cautiously to invertebrates like insects (Barrett and Fischer, 2023). Still, it is an improvement on the Five Freedoms framework, so we rely on it here.

3 Yellow mealworm biology

Understanding YM biology is essential for contextualizing potential industry-relevant welfare concerns. YM belong to the generally detritivorous, globally-distributed beetle family Tenebrionidae. *T. molitor* has a long history of human coexistence as a stored grain pest (Dunford *et al.*, 2005). From a likely Mediterranean origin (Panagiotakopulu, 2001), YM have spread to most of the temperate regions of the world, while high temperatures prevent establishing in the tropics (Bousquet, 1990; Cotton, 1927; Majka *et al.*, 2008).

YML live within their feeding substrate and go through a variable number of molts (9-23, typically closer to 17-19) dependent on abiotic and nutritional conditions (Cotton, 1927, 1940; Hein, 1923; Ludwig, 1956; Morales-Ramos *et al.*, 2010; Park *et al.*, 2014; Ribiero *et al.*, 2018). The time period from egg hatching to pupation can take as little as 2-3 months, or as long as two years (Cotton, 1927; Finkel, 1948; Ribiero *et al.*, 2018; Tracey, 1958). Developing YML increase ~10-fold in body length (Park *et al.*, 2014).

YML generally consume decaying grains, but will also consume cereals, bread, meal scraps, feathers, dead insects, live conspecifics, decaying wood, and grains mixed with animal excrement (Alexander and Anderson, 2012; Cotton, 1940; Ramos-Elorduy *et al.*, 2002). YML are uniquely adapted to living in particularly dry substrates as they may absorb water vapor directly from the air through their rectal complex (O'Donnell and Machin, 1988), though they grow better when provided with water sources and may prefer damp substrates (Cotton and St. George, 1929).

Although YML tend to aggregate within their substrate, they are highly mobile due to their prolegs; therefore, they may leave the feeding substrate to search for new resources. YML may remain in their final-molt stage for a prolonged period, prior to the onset of pupation (e.g., metamorphosis; Cotton, 1927). Hormonal cues cause final-molt YML to stop feeding and enter a "wandering phase" (Jindra, 2019; Muramatsu *et al.*, 2008), where they look for an appropriate pupation location. YM pupae have reportedly been found on fertilizer, salt, soda ash, tobacco, ground black pepper, and other unusual substrates (Cotton, 1940). Pupation, where pupae can only move their posterior segments, lasts 6-20 days (Cotton, 1927; Ribiero *et al.*, 2018).

Adult beetles are ~1-2 cm in length (McConnell and Judge, 2018; Morales-Ramos *et al.*, 2010). Males and females are visually monomorphic (Bryning *et al.*, 2005; Tanaka *et al.*, 1986). Adult beetles do not fly, are nocturnal and photophobic (Balfour and Carmichael, 1928), and generally live in the substrates where they reproduce for ~1-6 months (Cotton, 1927; Morales-Ramos *et al.*, 2012; Ribiero *et al.*, 2018).

Mating first takes place around 3-5 days after eclosion (Cotton and St. George, 1927; Drnevich *et al.*, 2001) and attraction is mediated by pheromones (Bryning *et al.*, 2005; Carazo *et al.*, 2004; Hurd and Parry, 1991; Tanaka *et al.*, 1986). Male pheromonal attractiveness is affected by immune system function, parasite load, body condition, and inbreeding (Pölkki *et al.*, 2012; Rantala *et al.*, 2002, 2003; Worden *et al.*, 2000; but see Kivleniece *et*

al., 2010). The completion of courtship behaviors are important for mating (Boukouvala *et al.*, 2021; Font and Desfilis, 2003).

Females store the males' sperm internally (Happ and Happ, 1975) to continuously fertilize their eggs. Females remain sexually receptive throughout their adult lifetime, and may mate multiple times (Drnevich *et al.*, 2001; Worden and Parker, 2001), though fertility decreases after the first 2-5 weeks post-eclosion (Morales-Ramos *et al.*, 2012; Park *et al.*, 2012). Females lay between 50 and 600 eggs, with an average of ~250-300 eggs/female (Deruytter *et al.*, 2019; Dick, 1937; Ghaly and Alkoaik, 2009; Morales-Ramos *et al.*, 2012).

Eggs are white and oblong, and laid singly or in small clusters within the substrate where adults are living; sticky surface secretions cause eggs to become covered in the oviposition substrate. Hatching time depends strongly on temperature, varying from 4-19 days (Cotton, 1927).

4 Industry rearing and slaughter practices

YMs are popular for industrial rearing due to their rapid growth (bioconversion rates of 2.6-6.1, dependent on diets; Van Broekhoven *et al.*, 2015), high nutrient content, minimum breeding requirements, and ease of handling (Bednářová *et al.*, 2013; Ribiero *et al.*, 2018). YML are generally the final product, and may be sold alive or dead (Grau *et al.*, 2017; Hong *et al.*, 2020). While adult YMs have a higher protein content than YML (Rumpold and Schlüter, 2013), increased chitin and their noxious defensive secretions mean they are generally only used for breeding (Attygalle *et al.*, 1991; Belluco *et al.*, 2013; Krinsky, 2019; Finke, 2007; Liu and Zhao, 2019; though chickens will consume freeze-killed adult beetles, Zega Enterprises, 2020).

Industry rearing practices will vary widely depending on the scale of the facility (e.g. Kok, 2017; Zega Enterprises, 2020). Yellow mealworm mass-rearing is covered in detail in Zega Enterprises (2020) and Cortes Ortiz *et al.* (2016), so we only briefly review these practices below.

Larval rearing

Commercially-reared YMs are fed 70-85% carbohydrate diets (generally cereal bran, flours) sometimes with protein mixed in and/or fruits or vegetables supplied for moisture and micronutrients (reviewed in Berezina, 2017; Cortes Ortiz *et al.*, 2016; Hong *et al.*, 2020; Rumbos *et al.*, 2020). YML are reared in stackable crates (com-

monly: 65 L × 50 W × 15 H) as close together as ventilation and temperature constraints allow. Larvae may be reared in solid-bottom crates for 4-5 weeks, which can cause frass (excrement) to accumulate, before transfer to crates with screens that allow frass to fall through the bottom, which may reduce the presence of mites (Cortes Ortiz *et al.*, 2016).

Temperatures are generally between 25 and 30 °C. Relative humidity (RH) varies from 50 to 75% (Adamaki-Sotiraki *et al.*, 2022). Producers may rear *T. molitor* under 24-hour dark conditions. Although rooms are climate controlled, the distribution of equipment that monitors versus manages climate conditions means that crates will not experience uniform conditions (variance from 25-35 °C and 55-90% RH can be expected). Filtration is generally recommended to reduce contamination, pollution, and/or pests within the facility (Cortes Ortiz *et al.*, 2016).

Adult rearing, mating, and oviposition

In commercial facilities, the breeding stock often develops in crates kept separate from the stock destined to become product. Pupae may be collected by hand or with automated sieve machines. Pupal loss due to cannibalism is common given the long developmental period of last-instar larvae (Cortes Ortiz *et al.*, 2016; Morales-Ramos *et al.*, 2015).

Adults may be reared in solid-bottom or screened crates (Morales-Ramos *et al.*, 2012). Wheat bran is provided as a nutrition and oviposition substrate and water sources are required. Continuous water sources tend to attract mites and pathogens, but reduce labor costs. Low densities help minimize egg cannibalism (Cortes Ortiz *et al.*, 2016).

Many producers will rear adults in roughly age-matched groups, so that adults aged 3-8 weeks can be culled simultaneously (potentially via freezing in air) after reproductive efficiency peaks (Zega Enterprises, 2020). To synchronize larval ages throughout the long adult oviposition period, adults must either be moved to new crates every 3-5 days or the capture crate for hatched larvae must be replaced every 3-5 days.

Processing for slaughter and transport

YML destined for product are collected long before pupation, with some recommending collection at ~100 mg (though they may grow to more than 250 mg; Ghaly and Alkokaik, 2009).

Prior to slaughter, YML are subjected to three pre-processing steps: removal from the substrate, washing, and fasting. Late-stage YML are removed from the rear-

ing substrate using a sieve (automated vibrations, or manual; Morales-Ramos *et al.*, 2011a; Shuliy Machinery, n.d.; Spranghers *et al.*, 2021). As of 2016, most US facilities were manually sieving, but the industry continues to automate (Cortes Ortiz *et al.*, 2016).

Washing (agitation or dipping in water) removes frass and rearing substrate from the external surface of the larvae. Fasting, a period of starvation for 1-2 days prior to slaughter, clears the digestive systems of YML so less frass makes it into the final product (Spranghers *et al.*, 2021; van Huis, 2021). Neither washing nor fasting has been found to reduce microbial contamination or affect bacterial community composition in YML, suggesting these practices may be ineffectual at their stated goal (Wynants *et al.*, 2017).

YML are not typically stunned or anesthetized prior to slaughter. The most common slaughter methods include microwaving, drying in a convection oven, air freezing, shredding/grinding, and steam blanching or boiling (Bordiean *et al.*, 2020; Spranghers *et al.*, 2021; van Huis and Tomberlin, 2017). Often freezing, or blanching/boiling, will be used prior to drying (EFSA, 2015; Selaledi and Mabelebele, 2021; Sindermann *et al.*, 2021; Vandeweyer *et al.*, 2017b). Methods are chosen based on the facility's level of automation, if they need wet or dry processing for their final product, and their effects on product characteristics (Bordiean *et al.*, 2020). Adults that are culled prior to the end of their natural lifespan are reportedly frozen in air (Zega Enterprises, 2020).

Finally, YML may also be sold live to zoos, pet stores, direct to consumer, etc., for feeding to exotic/pet animals, in which case they must be transported. It is recommended to transport YML refrigerated (Spranghers *et al.*, 2021). However, they may be shipped live directly to consumers in containers without climate control, water, or food.

5 Current welfare concerns for yellow mealworms and recommendations

We reviewed the scientific literature for information on yellow mealworms and interspecific interactions (including predators, parasites, and pathogens), abiotic conditions (temperature, hydration, atmospheric gasses, lighting), nutrition (including pollutants and hormones), intraspecific concerns (genetics, morphological defects, cannibalism, density, mating and oviposition needs, and handling-associated stress), and slaughter and depopulation methods (including anesthesia and stunning). A detailed review of everything

known about each of these factors can be found in Supplementary File S1, as not all information reviewed was relevant to the welfare concerns discussed further in this paper. While these factors may not be welfare-relevant at this time, the information may still be relevant to producers or researchers looking to improve production goals or as the basis of future investigations into the species' biology and welfare recommended in section 7.

From this review, we were able to determine welfare concerns that may currently manifest in industry conditions and develop species-specific, practical recommendations for improving current concerns and provisioning for positive welfare.

A few general caveats inform all of the paper's recommendations:

1. Throughout the paper, we frequently refer to improved fitness or growth as possible indicators of welfare; we include these metrics because they are among the most commonly reported measures in papers on YM (alongside survival). Importantly, *improved production outcomes do not always guarantee better welfare* – and in some cases, they can be negatively correlated. Further research will be required to validate if improved fitness or growth indicates positive welfare in each case for YM. However, we can be most confident that welfare is improving when other measures, such as survival or behavioral expression, also improve alongside fitness or growth.
2. Most studies have been conducted at laboratory scales, not production scales. Scale can nonlinearly influence many aspects of insect survival and, plausibly, welfare (Scala *et al.*, 2020; Sørensen *et al.*, 2012). Therefore, producers should consider the influence of their facility's scale when implementing welfare recommendations.
3. Strain and population can affect many welfare-relevant variables. Producers should assess their own population's performance under these recommendations and make adjustments accordingly to continue to improve welfare (e.g. looking to further reduce mortality, decrease the incidence of behaviors like tonic immobility, etc.).
4. Many concerns interact – for instance, malnutrition, cannibalism, or stress can exacerbate disease in YM (Eilenberg *et al.*, 2015; Siva-Jothy and Thompson, 2002). Therefore, it is important to consider welfare holistically and strive to implement recommendations that improve welfare across all domains. It is also important to understand that addressing one welfare con-

cern (such as inappropriate hydration) could have unintended, and currently unstudied, impacts on another aspect of YM rearing (e.g. fungal disease incidence).

For each concern, we review the relevant biological information that motivates the concern, before recommending improvements to industry practices.

Concern 1. Inadequate nutrition for larvae and adults

Nutrition impacts development and survival in larval and adult YM. Insect diets may fail to provide adequate nutrition if they are too homogenous, often resulting in inappropriate macronutrient ratios. Suboptimal macronutrient ratios may also influence the prevalence of cannibalistic behaviors in insect larvae (e.g. Randall *et al.*, 2023; see more on cannibalistic behaviors in YM in Concern 4).

Wheat bran (13–18% protein, 56% carbohydrates; Prückler *et al.*, 2014) is the most common component of YML diets, containing all the necessary nutrients for growth but in suboptimal proportions (Ribiero *et al.*, 2018, Morales-Ramos *et al.*, 2010). Feeding on wheat bran-only diets results in significantly longer developmental time and lower survival rates for YML (Morales-Ramos *et al.*, 2010). Despite this, feeding YML wheat bran-only diets may be common practice in the industry (Deruytter *et al.*, 2020). Similarly, adults in industry conditions are generally reared on a diet of wheat bran, with the provisioning of supplemental protein sources infrequently reported.

Recommendation (domain 1: nutrition): provide supplementary protein (minimally 20% for YML and 40% for YM adults) with egg, soy, or yeast. YML have increased survival on a heterogeneous diet of cereal (e.g. wheat bran), a dedicated source of protein (e.g. soy protein, beer yeast) and a water source (often vegetables, e.g. potato). Providing a source of protein improves pupal weight gain, reduces development time, and significantly increases larval survival rates (van Broekhoven *et al.*, 2015; Oonincx *et al.*, 2015; Morales-Ramos *et al.*, 2013; Figure 2).

Larvae preferred 20% egg protein diets over diets of soy protein, or diets that mixed peanut/salmon/canola oil with soy and egg protein (Morales-Ramos *et al.*, 2013); however, the study did not test yeast, which is the most commonly reported source of protein provided for YML (Ribiero *et al.*, 2018). Macronutrients can have synergistic or antagonistic interactions: for instance, high fat diets performed the best for survival when combined with high amounts of protein, but the worst when com-

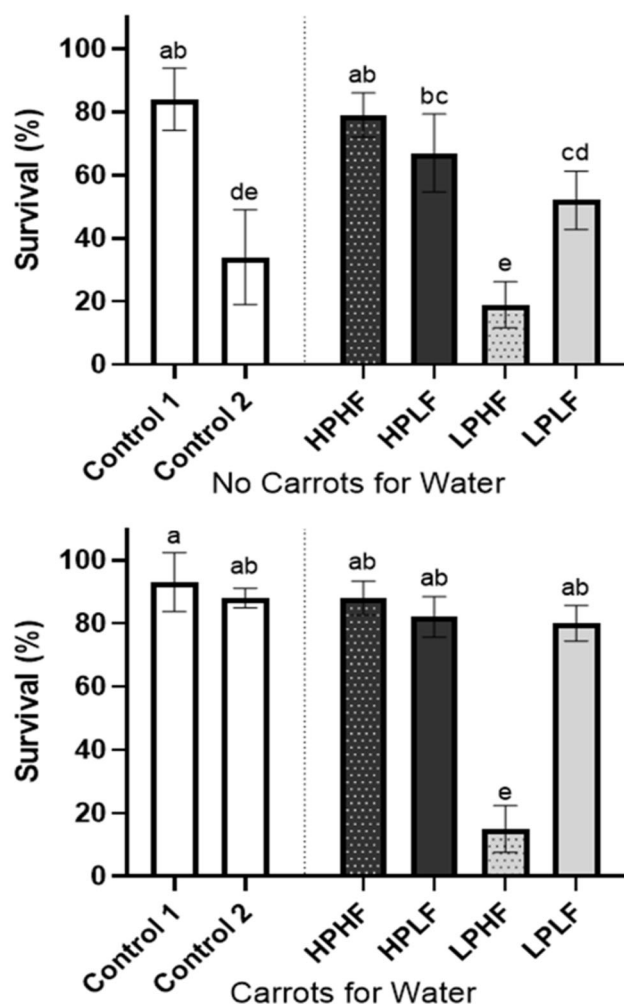


FIGURE 2 Effects of high/low protein and high/low fat diets on YML survival, with and without water sources (carrots). Data re-plotted from Table 3 in Oonincx *et al.* (2015); control diets 1 and 2 are not described, but are standard formulations obtained from two industry producers. HP = high protein; LP = low protein; HF = high fat; HL = low fat.

bined with low amounts of protein (Figure 2). Therefore, it is especially important to provide adequate supplemental protein when diets are higher in fats.

Finally, protein in the larval stage can also improve adult fecundity, with egg production increasing from 3–4 eggs/d to 6–7 eggs/d (Morales-Ramos *et al.*, 2013). Therefore, producers should strongly consider providing supplemental protein to YML destined to become breeding adults (as compared to product) for economic reasons.

As with larvae, dietary macronutrient balance (protein: carbohydrate, P:C) is a critical determinant of adult YM lifespan and reproduction. Average lifespan was the longest for males at P:C 1:1 and longest for females at P:C 1:1 and 1:5, while age-specific egg laying was highest at P:C 1:1 (Rho and Lee, 2016). Wheat-bran only diets will thus provide severe protein limitations for adults (P:C

of 1:3.1 – 1:4.3). Providing supplemental protein close to a 1:1 ratio will improve adult longevity and may also increase female reproduction.

Recommendation (domain 1: nutrition; domain 4: behavior): provide YML opportunities for nutrient self-selection using heterogeneous diets

YML are capable of nutrient self-selection to regulate their protein and carbohydrate intake. For instance, YML preferentially consume a 4:1 ratio of wheat bran: dry potato flakes. This ratio provided the shortest development time and was second best for population growth (Morales-Ramos *et al.*, 2011b), suggesting producers will still experience high productivity on YML-selected diets. Instead of typical, homogenous wheat-bran diets, industry can promote YML welfare by providing heterogeneous diets (e.g. supplementing with other substrates like potato flakes, yeast, etc.), allowing larvae to self-select optimal nutrition for their developmental stage.

In order for nutrient self-selection to occur, heterogeneous nutrients must be made accessible to the all animals. Small YML are less mobile than older larvae; producers should take care to distribute heterogeneous dietary resources well throughout the feed so all larvae will be able to take advantage of them. Finally, grinding of dietary resources (such as vegetable chunks or potato flakes) to a smaller particle size may be important – especially for smaller larvae – in order for them to be physically able to consume the provided macronutrients.

Adult yellow mealworms actively regulated their intake towards a slightly carbohydrate-biased ratio (males = P:C 1:1.54–1:1.64; females = P:C 1:1.3–1:1.36; Rho and Lee, 2016). However, this study used unusual feed ingredients (and did not test wheat bran), therefore results from this study should thus be interpreted cautiously. Still, this study supports the general principle that adults can self-select their macronutrient intake if provided with heterogeneous diets, allowing them to maximize their welfare by managing their own nutrition.

Concern 2. Inadequate hydration for yellow mealworm larvae

Poor hydration can reduce yellow mealworm survival, especially when individuals are at vulnerable life stages. As both eggs and adults die rapidly without appropriate humidity/water sources, appropriate provisioning of moisture for these life stages is a necessity for producers. However, as water sources can encourage fungal growth,

YML may be reared without access to a water source in industry conditions (reported in Adamaki-Sotiraki *et al.*, 2022).

YML are highly resistant to desiccation-induced mortality and may live for >150 days without water sources by relying on metabolic water production; this resistance is in part due to their ability to absorb moisture from the air through their rectal complex at certain RH (Buxton, 1930, 1932; Machin, 1975; Mellanby, 1958; Mewis and Ulrichs, 2001; Noble-Nesbitt, 1970; Schulz, 1930). Nevertheless, poor hydration can have lethal or sublethal impacts on their welfare.

Recommendation (domain 2: environment): provide larvae with 70-75% RH rearing environments

Larval body size and RH are positively correlated, with larvae in high RH (84%) conditions nearly two-fold the body mass of those in moderate RH (43%) conditions after three weeks of growth (Johnsen *et al.*, 2021). Larval growth rates are highest at humidities of 85-100% (Hardouin and Mahoux, 2003; Johnsen *et al.*, 2021). YML ceased activity and growth at low RH (e.g. 13-30%; Fraenkel, 1950; Murray, 1968) and pupation rates declined in low RH treatments (Graham *et al.*, 2000). The effects of poor hydration are strain-specific, producing variable effects on development time (Adamaki-Sotiraki *et al.*, 2022), larval weight, and adult longevity at 50% RH (Urs and Hopkins, 1973).

While high RH may be preferred for welfare, fungi, mites, etc. can proliferate in high-humidity conditions (or when water sources are provided; Frankel, 1950; Ghaly and Alkoik, 2009). High RH also increases sensitivity to high temperatures at all life stages (Punzo and Mutchmor, 1978, 1980). While 85-100% humidity may be best for larval growth rates at temperatures of 25 °C, 70-75% with supplemental water seems to provide optimal allowance for growth while reducing the twin welfare risks of thermal stress and disease that present at higher humidities.

Recommendation (domain 2: environment; domain 4: behavior): provide continuous access within 5 cm to water sources for all YML

Water sources that may be provided commercially include vegetables or fruits, hydrated gels, or sprays, which may be refreshed every few days (Cortes Ortiz *et al.*, 2016). Providing water sources increased growth rates and body masses of YML, and increased adult eclosion rates and longevity (at 50% RH; Mellanby and French, 1958; Murray, 1968; Urs and Hopkins, 1973; also, see decreased development times, 70% RH, Oonincx

et al., 2015). In addition, water sources increase larval survival (Oonincx *et al.*, 2015, and see trends in Urs and Hopkins, 1973; further, compare the same strains in Rumbos *et al.*, 2021 with Adamaki-Sotiraki *et al.*, 2022). Providing water sources is an important part of maximizing the survival and welfare benefits of appropriate larval nutrition (discussed in Concern 1) across strains (Figure 2; Oonincx *et al.*, 2015). The welfare benefits of providing supplemental water sources for larvae appear more pronounced at low-to-moderate RHs (Murray, 1968).

Beyond improved survival and growth, providing moisture sources allows individual YM to behaviorally regulate their water balance by directly ingesting water (Mellanby and French, 1958; Murray, 1968) or by changing their proximity to the higher RH microclimate near the water source (Barton-Browne, 1964; Ghaly and Alkoik, 2009). Dehydrated YML will move towards water, even when that requires migrating to the surface of a container, suggesting unmet hydration needs might be more stressful than photophobia (Martin *et al.*, 1976). YML have been observed seeking out damp areas in natural and laboratory conditions (Cotton and St. George, 1929; Howard, 1955; Urs and Hopkins, 1973). Water sources in the environment are an important part of environmental opportunity and choice for YML.

Provisioning water sources often requires significant labor for producers as they must be changed regularly to retain the moisture and keep crates clean; to save time, producers may thus place all the 'wet feed' in the center of the crate, instead of distributing it throughout the feed. Producers should always take care to distribute the wet feed throughout the crate, especially for smaller (<30 mg), less mobile larvae: decreases in growth rate were observed when wet feed was more than 5 cm from the larvae, with resulting larvae up to 150× smaller in body mass when found more than 15 cm from wet feed. As a result, Deruytter *et al.* (2020) recommend distributing wet feed so that it is always within 5 cm of the larvae for easy access.

Concern 3. Handling-induced stress, or other negative affective states, especially during pre-slaughter washing/fasting of YML

Farmed YM larvae and adults may be handled during the rearing process to:

1. Maintain the colony (e.g. remove dead individuals, age-match larval populations by moving adults, or change food/water sources)
2. Check health and development

3. Process and slaughter
4. Transport (see Concern 11)

Some handling is necessary to maintain high welfare conditions (e.g. changing substrates to provide quality nutrition or checking for signs of disease).

Recommendation (domain 2: environment): use red lights during handling

Light may be necessary during rearing or processing so that facility staff can safely complete their work. However, YM adults are nocturnal and photophobic (Balfour and Carmichael, 1928; Cloudsley-Thompson, 1953; Sheiman and Kreschenko, 2010). Larvae also demonstrate a strong preference for darkness (Howard, 1955; Loeb, 1905); YML bury themselves in the feeding substrate when exposed to light and will move away from introduced light even when no substrate is provided (Balfour and Carmichael, 1928; Howard, 1955). In fact, both larvae and adults avoid light more strongly than the lethal insect repellent, paradichlorobenzene (Howard, 1955).

YM have low spectral sensitivity to red, like most insects, and thus behave as if they are in the dark when maintained under red light conditions (Loeb, 1905; Wang *et al.*, 2022b; Yinon, 1970). Therefore using only red light (600–700 nm) will allow personnel to see while reducing light-associated stress for the insects. Using red lights is especially important whenever the insects are not provided with a substrate to hide in during the handling event (such as during fasting periods or depopulation events).

Recommendation (domain 2: environment; domain 5: mental): reduce the severity of disturbance and/or length of time of handling events during rearing

Handling events may be superficially similar to predator threats, such as being physically handled or the feeling of vibrations in the substrate; for this reason, handling may cause stress or possibly even fear-like states for insects. Predators are linked to stress hormone release in many insects (Cinel *et al.*, 2020), resulting in a suite of evolved anti-predator behavioral responses, such as tonic immobility (Humphreys and Ruxton, 2018).

Experimenters physically ‘jarring’ an experimental arena to cause vibrations resulted in tonic immobility in adult YM beetles (Krams *et al.*, 2013a). The presence of a nocturnal predator (whose activity caused vibrations in the substrate) increased hiding behaviors and tonic immobility in adult YM (Krams *et al.*, 2013b). Vibrations are linked to stress hormone levels in adult insects of

other taxa (Adamo and Baker, 2011), but this has not yet been studied in YM.

It is possible YM adults may acclimate to handling with more frequent handling events, as expressed by a reduced duration of tonic immobility (Howard, 1955). If further research supports acclimation as a response to frequent handling, it is possible producers may wish to establish routine, low-disturbance handling practices for young adult YM to reduce stress in response to handling events throughout their lifespan. In any case, frequency and duration of tonic immobility could be assessed as potential indicators of welfare status in farmed YM adults during handling events so producers can make informed decisions about how to reduce stress and fear-like states in their animals.

Increased stress in response to handling has not been studied in YML, though ‘violent locomotion’ is reported in response to rough handling (see discussion in Cloudsley-Thompson, 1953). Given that anti-predator adaptations are common in insect larvae as well as adults (e.g., Ramirez *et al.*, 2010; Sih, 1986), further research may find a useful behavioral indicator of welfare during handling by looking at YML anti-predator adaptations. Increased handling was hypothesized to reduce growth and survival in Deruytter *et al.* (2019); if this hypothesis is true, it supports minimizing both the frequency and severity of handling events during the rearing of YML, as larvae may not acclimate to disturbance.

Pre-slaughter handling may prove to be a particularly stressful experience for YML, as vibrating sieves may be used to separate YML from the feed. It's unclear if the vibrations produced by machinery are experienced similarly to the vibrations produced by predators that can cause stress in insects. Still, vibrations from nearby machinery were posited as a potential reason for slow larval growth in Li *et al.* (2015). At this time, producers should use whatever sieving method (manual or automatic) will minimize the length of time larvae spend experiencing vibrations, with future research aimed at determining the accumulated stress and/or fear experienced by the animals in response to these different operating procedures.

Recommendations (all domains): eliminate or reduce the length of time of pre-slaughter washing and fasting periods for YML. Provide burying substrates for YML during any fasting periods

Processing of larvae prior to slaughter may involve washing and fasting for ~48 hours, in theory to remove microbial contaminants (van Huis, 2021).

Alongside the stress that may be produced by agitation during washing, YML are not naturally found immersed in water; being submerged fully in water, even for short periods, represents a significant environmental challenge.

Further, washing and fasting removes the larvae from their feeding substrate for a prolonged period, restricting their opportunity to express natural burrowing behaviors. Behavioral observations of late-instar YML extracted from their feeding substrate show that they engage in a rapid searching behavior along the edge of the arena and, if provided with substrate, will immediately bury themselves again even if the substrate provided is not especially nutritive (Barrett, pers. comm). Given that wandering insect larvae are at high risk of predation in the wild (Aluja *et al.*, 2005), the tendency for YML to quickly re-bury themselves may be to avoid predation. Therefore, a lack of substrate may induce frustration through limited behavioral expression or stress/fear-like states due to the experienced greater risk of predation.

Continuing to wander for a prolonged period in search of a substrate represents an energetic cost to the animal and may indicate that the denial of burying opportunities fasting is stressful for YML. The increased energetic expenditure of unnatural wandering behaviors may make prolonged periods of fasting especially bad for YML welfare by inducing hunger/starvation or frustrating the desire to feed. Starved larvae had a cannibalism rate of 11%-37.2% compared to 1.7%-14.9% on a bran diet (Yang *et al.*, 2021a,b), suggesting fasting can induce changes in cannibalistic behavior that increase injury and reduce physical health.

Wynants *et al.* (2017) demonstrated that contamination levels in YML did not change as a result of either washing or fasting. Therefore, it may be possible to largely discontinue the use of these practices, improving production efficiency while simultaneously improving YML welfare. Of course, prior to discontinuing these practices, further tests with different rearing substrates, microbial loads, and environmental pollutants would be necessary to confirm the effect of removing washing and fasting steps on product safety and quality. However, producers should seek to eliminate or reduce the length of time spent on either practice.

If product safety constraints necessitate the use of long fasting periods, providing a non-nutritive burying substrate for YML during fasting periods would allow YML to engage in natural burying behaviors while avoiding any food consumption. Producers might look into inedible substances that are unlikely to cause injury and

can be easily cleaned (for example, small glass beads) until further research on the best non-nutritive burying substrates is available.

Concern 4. Injury, cannibalism, behavioral restriction, and/or other stressors driven by high stocking densities Stocking density can have multifaceted effects on animal welfare. Beyond direct effects of the proximity of so many animals (e.g. facilitating disease transmission or increased physical contact), too high densities can contribute to inadequate resource availability or accessibility and/or result in behavioral/physiological changes that negatively affect welfare.

When comparing the welfare of insects to vertebrate livestock, it is frequently suggested that rearing insects at high densities in intensive production systems is not likely to be detrimental to their welfare because insects are often found in nature at high densities (e.g. Dossey *et al.*, 2021; Erens *et al.*, 2012; IPIFF, 2019; and see discussion of this issue in Delvendahl *et al.*, 2022). First, it should be noted that natural conditions do not necessarily serve as a guide to high animal welfare. High densities in natural conditions may simply result from overutilization of scarce resources, not because insects' welfare is better under those conditions.

Second, it should be noted that almost no studies have documented the density of insects in natural conditions nor the prevalence of these densities in natural conditions (where we may also expect a bias towards the documentation of larger/high density populations that are easier to detect, as discussed in studies of other group-living animals: Markham *et al.*, 2015; Sharman and Dunbar, 1982). It is thus unclear what density insect producers should mimic, even if mimicking natural conditions was expected to produce the best outcome for welfare.

Finally, studies that robustly test the hypothesis that high density conditions improve, or harm, wild insect welfare are non-existent. However, high density conditions can substantially increase mortality due to density-dependent factors (e.g. risks of some diseases, resource scarcity, etc.; Lack, 1954). It is important to be critical of the claim that insect welfare is always improved under high density conditions until empirical evidence establishes the appropriate rearing densities for each species' welfare.

Nevertheless, let's suppose that YML *do* thrive under higher density conditions. Still, it remains the case that stocking densities in intensive production facilities could still easily exceed their optimal 'high' density.

Recommendation (domain 2: environment): rear YML at a maximum density of 0.5 larvae/cm³

Surprisingly few studies have directly assessed the impact of rearing density (areal or volumetric) on larval growth parameters, survival, or behavior under standardized abiotic conditions (Table 1). Increased density appears to decrease total weight gain, food consumption (particularly for later-instar larvae), and survival (Deruytter *et al.*, 2022; Morales-Ramos and Rojas, 2015; though see Deruytter and Coudron, 2019; Weaver and McFarlane, 1990).

Deruytter *et al.* (2022) found that optimal growth conditions for larvae were 0.5 larvae/cm³ – which was the lowest density tested, suggesting that optimal larval density for individual growth and welfare could be even lower. Rearing larvae at this density resulted in low total yield, increased space utilization, and feed waste. The authors thus recommend 1 larvae/cm³ to try to minimize the economic and ecological costs of low-density production. Similarly, Deruytter and Coudron (2019) recommend rearing larvae at higher densities that resulted in increased mortality in order to optimize yield. These studies suggest that there is a tension between maximizing yield and animal welfare, where the stocking densities that are optimal for the former are less than optimal for the latter.

Low stocking densities can also have underappreciated long-term benefits to welfare and economic productivity; for example, low-density housing may actually bolster immune responses through low-level exposure (Barnes and Siva-Jothy, 2000).

Recommendations (domain 3: physical health): to reduce injury and cannibalism, 1) separate larvae and pupae; 2) age-match larvae when rearing to reduce size variation; and 3) reduce densities at larval stages closest to pupation

For YML, high stocking densities can contribute to cannibalism – particularly of smaller larvae or pupae (which are particularly vulnerable due to their relative immobility; Ichikawa and Kurauchi, 2009). Cannibalism may not result in the immediate death of the cannibalized individual; instead, they may simply be injured but remain alive for hours (or days; Ichikawa and Kurauchi, 2009) prior to succumbing to hemolymph loss or putative septicemia (Weaver and McFarlane, 1990). Non-lethal wounds caused by cannibalism breach the cuticle, with holes that may be several mm in diameter (Weaver and McFarlane, 1990), and thus may also increase instances of disease (see Concern 5).

High larval rearing density may increase cannibalism, especially at later larval stages (Weaver and McFarlane, 1990; but see Zaelor and Kitthawee, 2018 where no effect was found). As density increased from 2 to 20 individuals/455 ml rearing jar, survival to pupation declined from 94% to 47.5%. Cannibalism was only observed at the higher densities (Weaver and McFarlane, 1990). In other cannibalistic insect larvae reared in artificial environments, regular crowding and malnutrition led to the evolution of increased cannibalistic behavior across generations (Vijendravarma *et al.*, 2013). Maintaining low rearing densities could assist the industry in avoiding accidental selection for increasingly cannibalistic, and thus economically unviable, strains.

Separating larvae and pupae (Morales-Ramos *et al.*, 2012), as well as ‘young’ and ‘old’ larvae (Zega Enterprises, 2020), is important to protect the most vulnerable groups. However, age-matched larvae may still cannibalize one another (Weaver and McFarlane, 1990).

In populations of YML being reared for breeding instead of as product, larvae will need to pupate. The presence of larval conspecifics can inhibit pupation in Tenebrionid beetles via vibrations, physical contact, or the presence of large amounts of frass containing juvenile hormone (Connat *et al.*, 1991; Schmialek, 1961; Tschinkel and Willson, 1971; Weaver and McFarlane, 1990). Therefore, rearing densities should be kept especially low in the larval stages prior to pupation to avoid possible behavioral restriction.

Recommendation (domain 2: environment; domain 5: mental): monitor temperature within crates of YML at high densities to ensure temperatures do not exceed 35–37 °C

The best ambient temperatures for rearing YM appears to be 25–28 °C, irrespective of humidity (Eberle *et al.*, 2022; Parsa *et al.*, 2023; Ribiero *et al.*, 2018), though growth rate was maximized with higher mortality at 31 °C (Bjørge *et al.*, 2018; but see increased development times at 30 °C vs 25 °C in Ludwig, 1956).

However, YML often aggregate within their feeding substrate; this may increase the local temperature far above ambient via metabolic heat production, making ambient temperature an unreliable indicator of larval thermal experience. For example, substrate temperatures greater than ambient by more than 10 °C have been recorded for late-stage larvae (Michal, 1931a; Morales-Ramos and Rojas, 2015). When combined with the insulating effect of the substrate, these localized hot spots can be lethal at high densities (e.g. in-crate temperature of 43.9 °C; Deruytter *et al.*, 2022). Producers

TABLE 1 Effects of different volumetric or areal YML stocking densities under controlled abiotic conditions

Density	Age	Abiotic conditions	Effects	Reference
1-20/400 ml*	Reared from 1 day old through adulthood	30 °C 55% RH 14:10 L:D Water provided (soaked cotton) Diet: wheat bran and whole wheat flour Food/larvae differed in treatments (400 ml food provided at all densities)	Increased body mass and survival unchanged with increased density at 1 month; female adults have higher body mass in low density trials; survival to pupation and adulthood declines with density	Weaver and McFarlane, 1990
44-351/dm ²	4th-6th instar larvae, reared for 12 weeks	27 °C 75% RH 0:24 L:D No water provided Diet: wheat bran (80%), dry potato, dry egg whites, and soy protein (20%) 34 mg food/larvae	Weight gain and food consumed decreased with increased density; no effects noted on survival or cannibalism	Morales-Ramos and Rojas, 2015
7.38 or 14/cm ²	12th-13th instar larvae, reared for 27 days	25 °C 70-80% RH 12:12 L:D Water provided (fresh ivy gourd leaves) Diet: wheat bran Feed given at 60% larval weight <i>Larvae were fasted every 3 days, for 3 days</i>	No difference in growth rate; 100% survival and no cannibalism in either condition	Zaelor and Kithawee, 2018
100-700/2064 cm ³ (e.g., 0.05-0.34 cm ³)	40 days after hatching, reared until pupation	25 °C 65% RH No photoperiod data Water provided (fresh fruits and vegetables, water sprayed on towel cover) Diet: wheat bran, 'livestock feed' Food/larvae differed in treatments (300 g food provided at all densities)	No effect of larval density on pupal weight	Park <i>et al.</i> , 2012

TABLE 1 (Continued)

Density	Age	Abiotic conditions	Effects	Reference
0.5-8 cm ³ , with areal varied from 0.5-64 cm ²	3-4 week old larvae, reared for 9 weeks [†]	27 °C 60% RH 0:24 L:D Water provided ('wet feed' agar gel) Diet: wheat bran Food/larvae differed in treatments (1-8 cm food provided to generate different volumetric densities)	Highest density (8 larvae/cm ³) reduced survival (68% at 4 weeks, compared to >95% at 9 weeks in other treatments); lowest densities increased growth rate and final larval weight	Deruytter <i>et al.</i> , 2022
Area varied from 0.6-10.4 cm ² , with volumetric stable at 1 cm ³	3-4 week old larvae, reared for 6-9 weeks	27 °C 60% RH 0:24 L:D Water provided ('wet feed' agar gel) Diet: wheat bran Food/larvae same in treatments (1 cm ³ food provided per larvae)	In conditions where frass was not removed, and substrate added, weekly: decreased survival with density; weight increased, then decreased, with density. In conditions where frass was removed, and substrate added, weekly: no change in survival with increased density; increased weight with increased density	Deruytter and Coudron, 2019
1-16/78.5 cm ²	age-matched, no data provided	No temperature data No RH data No photoperiod data Water provided (damp filter paper) Diet: wheat bran Food/larvae differed in treatments (30 g food provided at all densities)	Delayed pupation with increased density	Tschinkel and Willson, 1971

* The study says larvae were reared in 455 ml jars, and thus quotes density as individuals/455 ml. However, larvae are unlikely to float in the 55 ml of air above the 400 ml of provided substrate. So, here we provide the density as the # of larvae/unit substrate, which was 400 ml.

† Abstract lists incorrect densities; see Table 1 of study for correct densities.

↑ Deruytter *et al.* (2022) write "the high-density crates (8 larvae/cm³) were harvested after 4 weeks, as larvae health issues prohibited further continuation."

using high densities should monitor in-crate temperatures, and not ambient temperatures, to ensure YML are not overheating. Ambient temperature can be dropped, or larval density reduced, if in-crate temperatures are too high. 35–37 °C is a safe maximum temperature in the substrate, as critical thermal maxima for YML are between 40 and 44 °C (Ribiero *et al.*, 2018; Stevens *et al.*, 2010). Further, changing substrate textures and crate materials may allow for more rapid heat dissipation, protecting larvae from overheating.

Rapid increases in temperature, such as localized hot spots, are bad for survival and welfare. However, the prevalence of mortality from overheating is unknown (but see sublethal effects, Morales-Ramos and Rojas, 2015). Reducing thermal variability may be good for both productivity and welfare, due to the physiological costs imposed by large fluctuations in environmental temperature (Arias *et al.*, 2011). However, a small amount of thermal fluctuation may provide hormetic benefits (Colinet *et al.*, 2015). Finally, by providing a thermal gradient from the hot spot to the ambient-temperature edge of the crate, individual YML will have the opportunity to choose the microclimate that best suits their needs or preferences – assuming densities are low enough that they can easily and rapidly access all parts of the crate.

Concern 5. Diseases (particularly, caused by fungi and protists) and parasites

YM are host to a number of endoparasites including viruses, bacteria, protists, microsporidia, and fungi. We briefly review these known microbial pathogens, and major invertebrate parasites, in Table 2 (and see Slowik *et al.*, 2023). A detailed review of YM innate immunity and all these interspecific interactions can be found in the text of Supplementary File S1. This information may be useful to producers if practices can facilitate behaviors, cellular responses, or molecular pathways identified in preventing or overcoming pathogens.

Virulence can range from nearly undetectable to rapid symptom onset and death (Table 2). Both the severity and duration of welfare impacts, prior to death, will be relevant to the insect's welfare. Endoparasites may influence an insect's welfare by restricting their behavior (e.g. movement, feeding, or memory deficits), inducing malnourishment, damaging or distending the body (Eilenberg *et al.*, 2018), or even causing mortality. These changes may result in a variety of negative affective states, such as hunger in response to malnourishment or pain in response to prolonged mortality.

Although not yet validated as welfare indicators for insects, reductions in growth or reproductive output may also suggest welfare concerns when a disease-causing agent is present. The sublethal welfare impacts of most parasites and pathogens have not been studied; we thus are most likely to underestimate the welfare concerns they present without additional research.

To best understand the welfare of farmed insects it is important to consider both detrimental and beneficial relationships (e.g. YML glean benefits from microbiota; Lecocq *et al.*, 2021; and see Slowik *et al.*, 2023). Most of our knowledge of intraspecific interactions centers on the larval stage, but we have noted effects on adults, where studied in Table 2. It is unclear if research attention bias or reduced incidences of disease underlies the difference in literature on adults vs larvae.

Recommendation (domain 3: physical health): regularly monitor populations for disease and immune status. Asymptomatic or mildly symptomatic viral infections may go largely undetected in farmed insects until stressors instigate more widespread viral transmission or higher in-host replication, leading to epidemics (Fuxa *et al.*, 1999; Weissman *et al.*, 2012). Further, many pathogenic bacteria can persist outside of their host for at least a week (Dupriez *et al.*, 2022; Eilenberg *et al.*, 2015), allowing infections to periodically flare up when insects are stressed or facing multiple immune challenges. Regularly monitoring insects for mild signs of disease (behavioral changes like sluggishness, cuticular melanization or coloration differences, etc.) is the first step producers can take in spotting and preventing epidemics. Further, monitoring for symptoms can allow producers to depopulate diseased populations earlier in the disease's progression (see Concern 6) – avoiding prolonged suffering for the animals.

In addition to taking preventative measures, producers may also monitor the “immune status” of their colonies by assaying blood-cell counts (haemocyte number), immune system responses such as antibacterial or phenoloxidase (PO) activity in the hemolymph (Laughton and Siva Jothy, 2011), or even using cuticular color as a coarse measure of pathogen resistance (Armitage and Siva Jothy, 2005; and other detection methods are described in Slowik *et al.*, 2023). Rather than detecting a specific pathogen, this kind of monitoring would provide producers an index of the immune status of their colonies (Armitage and Siva Jothy, 2005). Low-level infections may go undetected for long periods of time until conditions such as crowding, high humidity, or weakened immune function facilitate outbreaks.

TABLE 2 Detrimental microbes (viruses, bacteria, fungi, and protists) and parasites of YM, prevalence, method of transmission, and symptoms associated with infection

Category (commonly used names)	Family: Genus species	Symptoms (in YM unless specified), prevalence, and method of transmission	References
Virus (parvo-viruses)	Parvoviridae: <i>Densovirus</i>	<i>Symptoms</i> : Very low levels of viral replication (PmergDNV). <i>Prevalence</i> : Little evidence as significant agents of disease for YML (AdDNV or PmergDNV).	Bertola and Mutinelli, 2021; La Fauce and Owens, 2008; Szelei <i>et al.</i> , 2011
Virus (iridescent Iridoviridae: <i>Iridovirus</i> viruses; IIVs)		<i>Symptoms</i> : iridescent hue of infected larvae, pupae, or adults due to viral replication in epidermis and fat body; sluggishness; patent infections nearly always lethal but can take several weeks. Sublethal infections that do not result in widespread iridescence likely affect growth and reproduction. <i>Prevalence</i> : Common and widespread agent of disease. <i>Transmission</i> : cannibalism, vertical transmission, or any cuticle-wounding activity; possibly ingestion, though less likely.	Bertola and Mutinelli, 2021; Black <i>et al.</i> , 1981; Carter, 1974; Genger <i>et al.</i> , 2020; Ince <i>et al.</i> , 2018; Kelly <i>et al.</i> , 1979; Williams, 1998, 2008; Williams <i>et al.</i> , 2005
Bacteria (Bt)	Bacillaceae: <i>Bacillus thuringiensis</i>	<i>Symptoms</i> : Cry proteins rupture the mid-gut epithelial cells, causing rapid death. Alternately, a slow death (averaging 30–40 days) by 1) causing gut paralysis and starvation, or 2) allowing bacteria to colonize the hemocoel, producing a severe septicemia. Reduced feeding behavior observed when infected. <i>Transmission</i> : ingestion of toxins; via cuticular wounds.	Broderick <i>et al.</i> , 2006; de Maagd <i>et al.</i> , 2001; Fabrick <i>et al.</i> , 2009; Graf, 2011; Hilbeck <i>et al.</i> , 2018; Oppert <i>et al.</i> , 2011; Raymond <i>et al.</i> , 2010; Zanchiet <i>et al.</i> , 2020*
Bacteria	Yersiniaceae: <i>Serratia marcescens</i>	<i>Symptoms</i> : low virulence, small increases in YML mortality. <i>Transmission</i> : ingestion; can persist in both feces and fed for over a week.	Dupriez <i>et al.</i> , 2022
Bacteria	Pseudomonadaceae: <i>Pseudomonas aeruginosa</i>	<i>Symptoms</i> : may increase cannibalistic behavior (as in giant mealworm larvae) <i>Prevalence</i> : detected in YM from commercial rearing facilities.	Marciel-Vergara <i>et al.</i> , 2018; Osimani <i>et al.</i> , 2018
Bacteria	Aeromonadaceae: <i>Aeromonas hydrophila</i>	<i>Symptoms</i> : none reported. <i>Prevalence</i> : not detected in farmed populations to date (e.g., Garofalo <i>et al.</i> 2017; NVWA 2014; Vanderweyer <i>et al.</i> 2017).	Eilenberg <i>et al.</i> , 2015

TABLE 2 (Continued)

Category (commonly used names)	Family: Genus species	Symptoms (in YML unless specified), prevalence, and method of transmission	References
Bacteria	Enterobac-teriaceae: <i>Salmonella</i> spp.	<i>Symptoms</i> : none reported. <i>Prevalence</i> : not detected in farmed populations to date (e.g. Garofalo <i>et al.</i> , 2017; NVWA, 2014; Vanderweyer <i>et al.</i> , 2017) or YM inhabiting chicken farms.	Jensen <i>et al.</i> , 2020; Krinsky, 2019; Wynants <i>et al.</i> , 2019
Protist (crypto)	Cryptosporidiidae: <i>Cryptosporidium</i> spp.	<i>Symptoms</i> : reproduce in the gastrointestinal tract of YML, often reaching very high densities and filling the host gut <i>Prevalence</i> : contemporary survey of pet food stores and small-scale rearing facilities in central Europe reported from 12% of facilities <i>Transmission</i> : cannibalism, ingestion of spores excreted into feed	Gałecki and Sokoł, 2019; Harry, 1967; Valigurová, 2007
Protist (Gregarines)	Gregarinidae: <i>Gregarina</i> spp.	<i>Symptoms</i> : reproduce in the gastrointestinal tract of YML, often reaching very high densities and filling the host gut. High densities could reduce longevity, slow movement, increase susceptibility to other diseases (but more data are needed). Generally, sublethal at low densities (though nutrient stress could induce negative effects via malnutrition); may even sometimes be beneficial to insects. <i>Prevalence</i> : contemporary survey of pet food stores and small-scale rearing facilities in central Europe reported from 33 % of facilities; high prevalence in natural populations <i>Transmission</i> : cannibalism, ingestion of spores excreted into feed	Arcila and Meunier, 2020; Eilenberg <i>et al.</i> , 2015; Gałecki and Sokoł, 2019; Harry, 1967; Koura and Kamel, 1993; Rodriguez <i>et al.</i> , 2007; Valigurová, 2007
Protist	Amoebozoa (Phylum)	<i>Prevalence</i> : rare.	Slowik <i>et al.</i> , 2023
Protist	Coccidia (Subclass)	<i>Prevalence</i> : rare.	Slowik <i>et al.</i> , 2023
Protist	unknown	<i>Symptoms</i> : non-fatal infection in 30-60% of population; localized, fatal outbreaks <i>Transmission</i> : cannibalism, ingestion of spores excreted into feed	Huger, 1967

TABLE 2 (Continued)

Category (commonly used names)	Family: Genus species	Symptoms (in YML unless specified), prevalence, and method of transmission	References
Fungi	Cordycipitaceae: <i>Beauveria bassiana</i>	<i>Symptoms</i> : Severe mortality in both adult and larval YML. <i>Prevalence</i> : Widespread in natural populations, reported from commercial operations. <i>Transmission</i> : breach the host cuticle using enzymes that weaken cuticular integrity and mechanical pressure to penetrate into the hemolymph	Aw and Hue, 2017; Bidochka and Kharchatourians, 1994; Donatti <i>et al.</i> , 2008; Eilenberg <i>et al.</i> , 2015; Eski and Murat, 2022; Lestari and Rao, 2016; Praprottnik <i>et al.</i> , 2021; Slowik <i>et al.</i> , 2023
Fungi	Clavicipitaceae: <i>Metarhizium</i> spp.	<i>Symptoms</i> : Mortality. Sub-lethal effects on adult reproduction (fewer eggs/reduced hatching when infected males mated with healthy females). <i>Prevalence</i> : Widespread in natural populations. <i>Transmission</i> : breach the host cuticle using enzymes that weaken cuticular integrity and mechanical pressure to penetrate into the hemolymph	Aw and Hue, 2017; Bidochka and Kharchatourians, 1994; Castillo-Perez <i>et al.</i> , 2022; Donatti <i>et al.</i> , 2008; Eski and Murat, 2022; Lestari and Rao, 2016; Praprottnik <i>et al.</i> , 2021; Slowik <i>et al.</i> , 2023
Fungi	Saccharomycetaceae: <i>Candida</i> spp.	<i>Symptoms</i> : Mortality (but only tested using direct injection).	Canteri de Souza <i>et al.</i> , 2015
Fungi	Nectriaceae: <i>Fusarium</i> spp.	<i>Symptoms</i> : dependent on mycotoxins produced; see Tables 3 and 4	See Tables 3 and 4
Microsporidia	Nosematidae: <i>Paranosema whitei</i> (previously, <i>Nosema whitei</i>)	<i>Symptoms</i> : none reported. <i>Prevalence</i> : rare. Only second and third-instar YML susceptible to infection. <i>Transmission</i> : infected feed.	Fisher and Sanborn, 1962; Milner, 1973; Slowik <i>et al.</i> , 2023; Sokolova <i>et al.</i> , 2005
Microsporidia	unknown	<i>Symptoms</i> : none reported. <i>Prevalence</i> : 87% of the beetles in the infected culture carried microsporidia; however, populations with infections are rare.	Armitage and Siva-Jothy, 2005

TABLE 2 (Continued)

Category (commonly used names)	Family: Genus species	Symptoms (in YML unless specified), prevalence, and method of transmission	References
Nematodes	Heterorhabdidae: <i>Heterorhabditis</i> spp. (harbors bacterial symbionts: <i>Photorhabdus</i>); and Steinernem-atidae: <i>Steinernema</i> spp. (harbors bacterial symbionts: <i>Xenorhabdus</i>)	<i>Symptoms</i> : releases bacterial symbiont into host hemolymph; bacteria consume host resources and produce toxins that result in poisoning and death within 2-3 days. <i>Transmission</i> : enter host insects through the mouth, anus, or spiracles; broad host range, especially concerning in high density rearing environments. <i>Heterorhabditis</i> can also enter through wounds in the cuticle	Ciche, 2007; Cranshaw and Zimmerman, 2013; Ferreira and Malan, 2014; Shapiro-Ilan 2008; Shapiro-Ilan <i>et al.</i> , 2014; Susurluk, 2006
Nematodes	Various vertebrate intestinal nematode species	<i>Symptoms</i> : reduced adult female fecundity but increased longevity, lethargy, sensory issues, impaired defensive responses and slower learning in a T-maze (<i>Hymenolepis diminuta</i>). Others have no symptoms reported. <i>Prevalence</i> : rare. <i>Transmission</i> : possible contamination from nearby vertebrates; consumption of eggs (<i>H. diminuta</i>).	Blankespoor <i>et al.</i> , 1997; Gałęcki and Sokoł, 2019; Hurd <i>et al.</i> , 2001; Hurd and Arne, 1986; Hurd and Fogo, 1991; Makki <i>et al.</i> , 2011, 2017; Sheiman <i>et al.</i> , 2006
Mites	Acariformes (Superorder)	<i>Symptoms</i> : unknown; some species may be commensal (see reports of <i>Acarus siro</i>), or only detrimental in large numbers or when YML are injured. <i>Prevalence</i> : Greater than 46% of central European rearing facilities had Acariadae present.	Buxton, 1930; Gałęcki and Sokoł, 2019; Howard, 1955
Parasitic wasps	Hymenoptera (Order): <i>Scleroderma guani</i> ; <i>Chouioia cunea</i> ; <i>Palmistichus elaeis</i>	<i>Symptoms</i> : YM pupae are stung; venom is used to suppress immune responses; wasp larvae hatch inside the pupae and consume it alive, taking an average of 12.9 days.	He <i>et al.</i> , 2006; Wu <i>et al.</i> , 2020; Zhu <i>et al.</i> , 2014

* Most research on Cry toxins and GMOs, not on direct *B. thuringiensis* infection.

It is thus especially important to monitor for disease-causing agents regularly, even when symptomatic individuals are not present.

Recommendation (domain 3: physical health): avoid feeding waste as feed to YML without appropriate pre-treatment to reduce disease

While YML will eat discarded food scraps, farmed insects fed kitchen discards and locally collected feed sources showed higher infection rates of protozoan fungal pathogens than those fed only fresh products or specialized feed (Galecki and Sokok, 2019). This suggests that integrating a waste-as-feed model in rearing YM will likely exacerbate disease outbreaks without appropriate pre-treatment (see next recommendation). Currently, the safest feed for YML appears to be fresh products and non-waste grains. Nevertheless, if waste is to be considered as feed, opting for organic waste may contribute to a safer diet by minimizing pesticide levels (though this remains to be tested). Additionally, waste sourced from a single location (as compared to, say, residential waste streams) would reduce the possible entry points for diseases.

Recommendation (domain 3: physical health): heat-treat feed prior to introducing YM

Heating feed to 60–90 °C prior to larval introduction significantly reduces *Gregarina* spore viability and subsequent infections of YML (MacDougall, 1942) and can inactivate Cry toxins from *B. thuringiensis* (Ujváry, 2010). Heat treatment may also be expected to reduce viral, bacterial, and fungal diseases and the presence of parasites. Importantly, heat treatment will also kill beneficial microorganisms; producers may wish to inoculate heat-treated feed with beneficial microbes (see next recommendation).

Recommendation (domain 3: physical health): consider providing probiotics for YML

More research is needed on which probiotics are most beneficial for YML welfare. However, *Pediococcus pentosaceus*, related to a genus of lactic acid bacteria often associated with beneficial roles in vertebrate intestinal health (Dempsey and Corr, 2022), along with three other species used as probiotics in vertebrates, *Bacillus subtilis*, *B. toyonensis*, and *Enterococcus faecalis*, significantly increased larval weight gain when provided as food additives (Lecocq *et al.*, 2021; Rizou *et al.*, 2022). The addition of these putative probiotic species to food also reduced the presence of bacteria in the order Enterobacteriales, which contains a number of known bac-

terial pathogens. *P. pentosaceus* isolates inhibited the growth of pathogenic bacteria (Lecocq *et al.*, 2021), and improved larval growth and survival of YML infected with the pathogenic fungus *Metarhizium brunneum* (Dahal *et al.*, 2022), suggesting this species might have antibiotic or antifungal properties.

Supplementing YML diets with probiotics may, through direct competition or by increasing nutrient availability, mitigate or prevent detrimental effects of pathogens, including any pain/discomfort and/or death associated with infections. If YML in mass-reared facilities are generally provided food without bacterial supplementation, even seemingly healthy individuals may be in substandard health. This suggests there is room to improve larval welfare, survival, weight gain, and health through probiotic supplementation.

Recommendation (domain 3: physical health): follow best practice guidelines for hygiene

Disease outbreaks represent a major animal welfare and economic productivity concern for the IAFF industry; accordingly, there are many sources already available that detail other farmed insect disease prevention measures, especially in relation to production facility infrastructure and hygiene practices. We advise all producers to carefully read and incorporate the standard hygiene practices recommended in Eilenberg *et al.* (2015, 2018), Joosten *et al.* (2020), Slowik *et al.* (2023), and International Platform of Insects as Food and Feed (2022).

Recommendation (domain 3: physical health): consider immune priming, through intentional introduction or low stocking densities, to promote high colony health

While bacterial or fungal overgrowth are problematic, a sterile environment should not be the aim as the YM immune system is bolstered by low-level exposure to immune challenges. Producers may take steps to bolster the immune status of their YM colonies through immune priming via logistic choices (such as low stocking densities; Barnes and Siva-Jothy, 2000) or through intentional immune priming (such as mild exposure to pathogens, toxins, etc.). For example, preexposure of YML to bacterial toxins appears to induce a long-term antimicrobial response that enhances survival (Moret and Siva Jothy, 2003). While immune priming requires exposing YM to mild stress, the short term welfare challenge may be outweighed by the long term welfare benefits of improved immune function.

Immune priming can also be transgenerational such that exposure in the adult stage induces greater immune response in offspring (Grau *et al.*, 2017; YM evidence

reviewed in Kojour *et al.*, 2022). This may be beneficial from a welfare perspective, as exposing a smaller number of breeding adults to a mild stressor produces a large number of offspring who have not experienced that stressor but are nevertheless better protected against disease.

Concern 6. Inhumane slaughter of YML and inhumane depopulation of YML or adults

Welfare during slaughter may represent a challenge to multiple functional domains, particularly physical health, environment, and behavior, with downstream effects on the mental domain (e.g. potentially causing pain, distress, etc.). No regulations exist to guide insect producers in implementing humane slaughter standard operating procedures (SOPs; Bear, 2019), and no data exist to inform regulators about the welfare implications of different slaughter methods. YML are currently slaughtered using shredding/grinding, blanching/boiling, microwaving, drying in a convection oven, and air freezing (Bordiean *et al.*, 2020; Spranghers *et al.*, 2021; van Huis and Tomberlin, 2017).

Disease, improper abiotic factors, inbreeding and other conditions may cause producers to need to remove crates of developing YML from the rearing facility and depopulate them, prior to the intended point of slaughter (or to cull crates of pupae or adults). Further, adults are generally depopulated when their fertility declines before the end of their lifespan, via freezing in air (Zega Enterprises, 2020).

Recommendations (domain 2: environment; domain 3: health; domain 5: mental): use slaughter and depopulation methods that are more likely to be instantaneous. Develop SOPs that reduce the time-to-death of the method used

Generally, slaughter methods that take less than one second are expected to be more humane in the absence of anesthetics or stunning, as the rapidity of death will reduce the risk of pain (IPIFF, 2019) or prolonged experiences of environmental challenge/injury. Barrett *et al.* (2022) describe in-depth the lethal progression of different slaughter methods, and rank each method as more or less likely to be humane. The information in Barrett *et al.* (2022) on is expected to be broadly accurate for YML, so we do not revisit it here. Blanching/boiling, shredding, and freezing (in liquid nitrogen or, with some debate, air) have generally been recommended as the most humane methods for killing insects (Barrett *et al.*, 2022; Erens *et al.*, 2012; Sindermann *et al.*, 2021; Spranghers *et al.*, 2021).

The rapidity of death using these methods will depend on the exact SOP used and the species' physiology (Barrett *et al.*, 2022). Consider the case of blanching (or boiling), which has the potential to be an effective, instantaneous method for slaughtering YML while also protecting product quality (Mancini *et al.*, 2019; Sindermann *et al.*, 2021). Blanching is low-cost, reduces microbial contamination and lipid oxidation rates, reduces browning, and increases the efficiency of the drying process (Larouche *et al.*, 2019; Mancini *et al.*, 2019; Seho *et al.*, 2021; Sindermann *et al.*, 2021; Rumpold *et al.*, 2017; Vandeweyer *et al.*, 2017a,b). These safety and economic benefits, alongside the welfare benefits resulting from an instantaneous death, mean that energy-efficient blanching SOPs should be a top priority for supporting high-welfare practice.

Blanching SOPs should be designed to guarantee instant death for all YML in a batch of a particular size (considering number of larvae, volume/shape of container, water:larvae ratio, etc.). While producers should also seek to minimize energy costs for sustainability purposes (e.g. by reducing the temperature of the water), a safety margin that ensures instantaneous death should be maintained. As both YM larvae and adults are highly mobile, SOPs should be developed that either (1) confine individuals, or (2) anesthetize or stun them, such that they cannot prolong their death through movement away from the instantaneous source of slaughter.

Regular slaughter equipment and/or protocols may not be appropriate to handle YM that are morphologically and/or physiologically distinct from slaughter-weight, late-instar larvae (e.g. smaller-bodied larvae may pass through regular grinding apparatuses largely unaltered). In addition, the conditions that cause producers to need to depopulate insects prior to slaughter could alter pre-processing in ways that change the effectiveness of slaughter methods. For example, fungal growth within feed substrates could make it difficult to separate YML from the substrate. Inclusion of substrate alongside YML or adults when using any temperature-based slaughter/depopulation procedures could then increase time-to-death, compared to animals separated from the substrate. Producers should thus develop specific protocols for depopulation under a diverse array of conditions and at different life stages.

Recommendations (domain 2: environmental challenge; domain 5: mental): avoid freezing in air as a method of anesthesia, depopulation, or slaughter. Freezing should never be used prior to heat-based methods of slaughter

Cold is the only anesthetic that producers mention they may employ prior to slaughter, assumedly in regular -18°C freezers, due to the popular claims that it 'gently puts insects to sleep' by reducing their metabolic rate (Bear, 2019). However, insects have nociceptors for, and unique behavioral responses to, noxious cold (e.g. fruit fly larvae; Turner *et al.* 2016), attempt to escape cold areas (Barrett, pers. comm.), and have prolonged behavioral and physiological alterations following recovery from cold anesthesia (Gooley and Gooley, 2021; MacMillan *et al.*, 2017). Therefore, it is plausible that cold may not be a high-welfare anesthetic for insects, as it may expose insects to noxious, stressful, and potentially pain-inducing stimuli prior to unconsciousness. For this reason, cold is not recommended as an anesthetic for invertebrates by veterinary specialists at this time (Cooper, 2001, 2011).

The period of time YM are exposed to possible pain from noxious cold lasts from the beginning of cold application to the point of chill coma (the temperature at which the nervous system of the insect is presumably inactivated, and therefore pain can no longer be experienced). YML chill coma temperatures are reported between 0.5 and 1.3°C at the fifth instar (Arias *et al.*, 2011). It can take several minutes for individually-frozen YML or adults to reach their chill coma temperature (see rates of cooling for YM in Constantinou and Cloudsley-Thompson, 1985). Because large numbers of aggregating insects produce metabolic heat, cooling times to the point of chill coma will invariably take much longer for mass-frozen individuals in the industry.

Further, although chilling in a -18°C freezer will always serve to render YM unconscious eventually, the process may not always successfully kill YML as a method of slaughter or depopulation. YML supercooling (freezing; SCP) points are as variable as -1.5 to -24.3°C , dependent on body mass/developmental stage (mean = -12.4°C). -24.3°C is lower than typical freezer temperatures (-18°C) and thus some YML may plausibly survive through freezing into the next processing phase (Constantinou and Cloudsley-Thompson, 1985; Johnston and Lee, 1990; though, larvae may die before their SCP due to other chill-induced injuries, Andreadis and Athanassiou, 2017). If larvae are slaughtered at the recommended 100 mg, their mean SCP will be around -14.4°C (Johnston and Lee, 1990). However, age and rearing

conditions can affect the SCP (see Supplementary File S1 for more information), meaning that depopulation efforts need to consider these variables when trying to depopulate or slaughter YML using this method.

Finally, cold is a particularly poor choice of anesthetic prior to any heating-based slaughter methods: chill coma will be reversed during the lethal progression to critical thermal maxima and, by starting at a cooler temperature, the animal will experience an even longer time-to-death than if they started out at room temperature. If cold is to be employed, it should precede continued freezing, or grinding/asphyxiation performed at temperatures well below the chill coma temperature.

Recommendation (domain 5: mental): develop stunning or anesthesia protocols prior to non-instantaneous depopulation or slaughter
Producers intent on non-instantaneous depopulation or slaughter methods should consider anesthetics, or electrical stunning, prior to slaughter. Barrett *et al.* (2022) covers current anesthetic options for insects; therefore, we only briefly review this topic here. Inhalation anesthetics, such as CO_2 or halogenated ethers, can be used to knock out insects prior to slaughter, and may thus benefit insect welfare by reducing possible experiences of pain (in the case of CO_2 , without any effect on product quality; Zhen *et al.*, 2020).

However more data are needed on the welfare impacts of CO_2 anesthesia itself, as many veterinary specialists do not recommend CO_2 as a humane method of insect euthanasia (Gunkel and Lewbart, 2007; Murray, 2012; and see Gooley and Gooley, 2021; MacMillan *et al.*, 2017). Additionally, the best protocols for employing inhalation anesthetics for farmed insects *en masse* must be studied; anesthetics given at incorrect doses can be ineffective or even harmful for animals (Marquardt *et al.*, 2018; Turner *et al.*, 2020). The halogenated ether isoflurane may prove best for insect welfare (Gooley and Gooley, 2021; MacMillan *et al.*, 2017), but its effects on product quality and safety are unknown.

Electrical stunning prior to slaughter may be used to incapacitate animals' nervous systems, including invertebrates (within 0.5 seconds; Neil and Thompson, 2012) at industry scales with low error rates (Roth and Øines, 2010; Roth and Grimsbø, 2013). Short applications of current can be used to stun, but longer applications can actually serve as a method of humane slaughter during the period of insensibility – allowing producers that adopt electrical stunning to move straight from stunning equipment to processing their product (Conte *et al.*, 2021).

No studies have tested electrical stunning at scale for farmed insects. However, smaller electrical devices have been used to stun insects for field collection (Packer and Brady, 1990) and pest management/monitoring (Harris, 2006) purposes. Given similarity in nervous system function between insects and other animals, electrical stunning devices will work on insects – they simply need to be designed for use at industry-relevant scales.

Importantly, insects can sense electrical fields and often find them aversive, seeking to avoid contact (England and Robert, 2021). In adult cockroaches, electrostatic deflection of mechanoreceptors on the antennae appears to be important for sensing electrical fields and mediating (generally) avoidance behaviors. How/if insects without antennae, like mealworms might sense electrical fields is currently unknown. The design of electrical stunning equipment should take into account the adverse effects of electrical fields on insects pre-stunning, by reducing any time spent in proximity to a field prior to the actual stunning experience.

Concern 7. Poor ventilation during rearing resulting in hypoxia for YML

Environmental oxygen levels are typically around 21% (normoxia). Hypoxia (10%) reduces growth rates, increases mortality, and causes developmental abnormalities in YML, including the inability to develop pigment which may increase susceptibility to disease (Clark and Cristofalo, 1961; Greenberg and Ar, 1996; Loudon, 1988). Hypoxia was significantly worse for survival than normoxia or hyperoxia (40%), with 20%, 96%, and 96% survival to pupation, respectively (Greenberg and Ar, 1996).

Mass-reared insects produce carbon dioxide (CO_2), ammonia (NH_3), methane (CH_4), and nitrous oxide (N_2O) gasses, though ammonia levels appear to be particularly low in *T. molitor* (insect gas emissions are partially dependent on feed characteristics, e.g. Lindberg *et al.*, 2022). The production of these gasses by large numbers of individuals in poorly ventilated spaces can create hypoxic conditions.

Recommendation (domain 2: environment): never keep crates of YML completely closed/sealed; ensure good air flow through crates, especially at ‘night’

Active, growing YML fed *ad libitum* are likely to produce the highest gas emissions (compared to adults, or inactive/unfed individuals; Oonincx *et al.*, 2010). The risk of hypoxia in poorly ventilated rearing containers is highest during ‘night’ (e.g. dark) conditions, when YML oxygen consumption and gas production are highest due

to their increased activity (Michal, 1931b; Oonincx *et al.*, 2010).

Closed systems can accumulate gasses quickly. A closed YML rearing system caused a 2.5× increase in average CO_2 levels, to 1000 ppm (with fluctuations in CO_2 concentrations from 1.25 to 15× that of the open environment). Increased CO_2 correlated with reduced larval growth (Li *et al.*, 2015). High levels of carbon dioxide (CO_2) can also cause insect spiracles to remain open, increasing water loss and desiccation stress (Nicolas and Sillans, 1989). For this reason, crates should never be kept completely sealed.

Even when crates are not sealed, farm-relevant conditions showed CO_2 levels reaching 1,500 ppm when larvae are reared at high densities (Deruytter and Coudron, 2019). Producers should, therefore, ensure good air flow in crates, especially at night and when rearing densities are high.

Recommendation (domain 2: environment; domain 4: behavior): provide appropriate light cycling for YML YM behaviors and the visual system are regulated by a circadian rhythm even when maintained at 0:24 or 24:0 light-dark (Erber, 1975; Fondacaro and Butz, 1970; Ro and Nilsson, 1993; Wada and Schneider, 1968; though see Cloudsley-Thompson, 1953; Howard, 1955). Photoperiods for rearing YM populations have included: 18:6, 16:8, 14:10, 12:12, 10:14, and 0:24 light-dark (reviewed in Eberle *et al.*, 2022; Ribiero *et al.*, 2018). Although further information is needed on the best photoperiods for rearing YML, providing periods of light and darkness may thus allow for natural periods of high and low activity to be expressed (Oonincx *et al.*, 2010). Therefore, minimally 0:24 and 24:0 cycles should not be used.

Appropriate light cycling will encourage reductions in gas output during the ‘day’, when activity is reduced; this will provide time for hypoxic conditions to be cleared out prior to the next ‘night’ period, eliminating extreme and potentially stressful fluctuations in oxygen levels (e.g. Li *et al.*, 2015).

Concern 8. Rearing YM adults without substrate on mesh

Producers may use steel mesh to separate adults from the underlying oviposition substrate, in order to prevent egg cannibalism. However, this also prevents adults from engaging in natural behaviors (namely, burying themselves within the substrate) and may expose photophobic adults to light (and potentially other stressors), presenting a welfare concern.

Recommendation (domain 4: behavior; domain 5: mental): rear adults with adequate access to oviposition substrates

Adults should have, minimally, enough oviposition substrate provided that they can bury into it and be completely covered, and to hide from any light sources or other stressors.

Concern 9. Inbreeding depression resulting in poor physical health

Inbreeding depression may occur during artificial selection processes (e.g. when effective population sizes of breeding adults are likely to be smaller), or accidentally whenever continuously breeding closed populations of captive insects (Eriksson and Picard, 2021; Gilchrist *et al.*, 2012). Inbreeding can result in reduced growth, morphological defects, decreased longevity, and reduced stress or disease resistance (reviewed in Jensen *et al.*, 2017), which may result in colony collapse and should thus be avoided for welfare and economic reasons. Inbreeding for even a single generation reduced adult YM mass and pre-adult survival, increased developmental time, and reduced resistance to *B. bassiana* fungi (Rantala *et al.*, 2011). Male pheromonal attractiveness is also affected negatively by inbreeding which may affect colony level reproductive output (Pölkki *et al.*, 2012).

Recommendation (domain 3: physical health): monitor the genetic health of reproductive stock populations and avoid inbreeding. If inbreeding occurs, reintroduce genetic diversity

Producers should take care to intentionally avoid inbreeding by monitoring the genetic health and diversity of their breeding population. If inbreeding occurs, genetic diversity can be reintroduced to reduce the negative welfare consequences through strain hybridization or crossing with wild populations (Jensen *et al.*, 2017).

Concern 10. Morphological defects in YM adults that restrict behavioral expression

Morphological defects in the adult life stage result from inappropriate development. The prevalence of morphological defects in YM populations is not well documented; one lab study reports 4% of adults are defectively formed (Errico *et al.*, 2021). Deformation in insects may result from genetics, disease, abiotic factors, or high larval rearing densities (possibly the result of consuming compounds with juvenile hormone activity found in the frass of conspecifics; Weaver and McFarlane, 1990). Producers should follow density (Concern 4), disease management (Concern 5), and genetics (Concern 9) rec-

ommendations, alongside the below recommendations specific to this concern, to manage the risk and impacts of morphological defects.

The welfare effects of deformation are not well understood. Some morphological deformations, such as a twisted wing, are unlikely to affect behavioral expression in adults. However, the retention of abdominal pupal characteristics will restrict normal locomotion, mating/oviposition, and potentially digestion. More serious defects may cause, or contribute to, mortality.

Recommendation (domain 2: environment): do not store YM pupae or larvae at cold temperatures (minimally, 10 °C or below)

Cold temperatures during development can result in morphological defects for later life stages. For example, storage of YM pupae at 4 °C resulted in 100% deformity of the wing or elytron of enclosed adults (Patterson and Duman, 2001; and see, Sömnez and Koç, 2019). Similarly, increasing the amount of time larvae were in cold storage at 4 °C increased mortality and adult deformation rates, with especially dramatic effects for young larvae (Sömnez, 2021). Producers should not expose larvae or pupae to temperatures below 10 °C.

Recommendation (domain 3: physical health; domain 4: behavioral restriction): monitor and humanely slaughter severely deformed adults

Eclosing adults should be monitored for deformations of the abdomen and humanely slaughtered whenever deformations are considered severe enough to restrict normal behavioral expression.

Concern 11. Inadequate nutrition and environmental conditions during shipping and rearing/storage of YML, sent direct to consumers or in pet stores

YML are transported live when sold to pet stores, zoos, or direct to consumers. The process of shipping, inappropriate storage practices, and neglect or inexperience by consumers and pet stores can result in starvation, desiccation stress, disease, cannibalism, and more.

Recommendation (domain 2: environment; domain 5: mental): do not ship live insects when climate conditions at the destination, or along the route, are not favorable

When shipped without climate control, YML may experience global thermal stress during heat waves or cold fronts, while on trucks or when left outside during shipping and receiving. Global thermal heat stress can produce pain-like responses in insect larvae (Chattopad-

hyay *et al.*, 2012); they also respond nocifensively to cold (Turner *et al.*, 2016). Producers should only ship live insects when climate conditions along the entire route are favorable to insect welfare, or with climate control (e.g. heat or cold packs, on climate-controlled trucks) to destinations where conditions on arrival are expected to be favorable.

Recommendation (domain 3: physical health; domain 5: mental): reduce handling-associated stress by marking packages as 'fragile' and using vibration-reducing shipping materials

Vibrations from transport vehicles may induce stress (see Concern 3) – inclusion of some packing materials, such as Styrofoam, may help to minimize vibrations. Additionally, rough handling by shipping personnel could induce stress or cause injury. Marking packages as fragile, with live animals inside, can encourage shipping personnel to treat the packages with greater care.

Recommendation (domain 2: environment; domain 4: behavior): do not refrigerate (<10 °C) any YML that weigh less than 30 mg during storage. Do not refrigerate older YML or pupae for prolonged periods (greater than two weeks)

YML or pupae may sometimes be stored refrigerated when shipped in order to extend the duration of time spent at that life stage. 30 mg may be the critical weight for cold storage of larvae at 10 °C, as smaller larvae saw increased mortality (Punzo and Mutchmor, 1980). Therefore, larvae less than 30 mg should never be stored at less than 10 °C.

98% of 2-3 month old larvae survived 120 days at 10 °C (63% RH). However, 13–19% of larvae did not survive to adulthood, compared to 1.5% of controls (Errico *et al.*, 2021). Reduced survival after 10 °C storage suggests being reared in these conditions for a prolonged period results in poor welfare, even for older larvae. Further, refrigeration results in behavioral restriction – reducing natural behaviors like feeding, locomotion, pupation, or eclosion, and causing morphological defects (see Concern 10). Producers should limit cold storage of larvae to two weeks – and provide cold storage removal dates on containers of YML sent to pet stores.

Recommendation (all domains): provide appropriate nutrition, hydration, and environments for YML during transport

YML are often shipped at high densities without water (and sometimes without feed); this may increase mor-

tality and instances of cannibalism. Producers should provide feed, water sources, and appropriate densities for larvae during shipping.

Recommendation (all domains): provide care instructions to pet stores and consumers to reduce neglect

Inadequate rearing conditions for YML in pet stores and by consumers, whether caused by ignorance or neglect, lead to reduced welfare and, often, significant mortality. YML are generally not provided with fresh feed or water sources in stores; they may be kept for prolonged periods in cold storage and at high densities. Very little monitoring or management of animal welfare may be provided by stores or consumers themselves.

Producers should provide pet stores and consumers with care instructions specifying how to provide adequate nutrition, hydration, abiotic conditions, and densities for the animals.

6 Future possible welfare concerns for yellow mealworms and recommendations

We also identified four future welfare concerns that may manifest as the industry scales but do not currently appear to be common practice.

Future concern 1. Feeding of plastics or polymers to YML, resulting in inadequate nutrition

YML may be fed plastics/polymers, including polystyrene, polyethylene, poly vinyl chloride, polyester polyurethane, vulcanized natural latex rubber (tire crumb), Styrofoam, and vulcanized butadiene-styrene elastomer (reviewed in Khan *et al.*, 2021; Yang *et al.*, 2018). Plastics/polymers may be fed with, or without, bran additives. Importantly, YM to be used as food and feed are unlikely to be reared primarily on these substrates – given concerns with consumer marketability, product quality, and health/safety. However, if this practice were to manifest at scale, it would represent a serious welfare concern.

Recommendation (domain 1: nutrition; domain 3: physical health): avoid feeding YML polymers/plastics. If YML are to be fed polymers/plastics, incorporate adequate dietary additives as a significant part of the overall nutrition provided

YML can survive and reproduce on some 100% polymer diets (Nukmal *et al.*, 2018) however slow growth rates (Billen *et al.*, 2020), increased metabolic stress

(Tsochatzis *et al.*, 2021), and higher mortality rates (Wang *et al.*, 2022a; Jin *et al.*, 2022, Yang *et al.*, 2021a), have been cited as reasons why large-scale rearing on polymers may be impractical as well as bad for animal welfare. Given trends of higher mortality and cannibalism, and the unnaturalness of polymers as feed leading to malnutrition, providing polymers to YML is not recommended. There is no known mechanism of pre-treating polymers to improve their safety for YML at this time. If plastics or polymers are to be fed to YML, the inclusion of bran additives can increase survival or growth/pupation rates (Lou *et al.*, 2021; Yang *et al.*, 2018) and reduce rates of cannibalism (Yang *et al.*, 2021b). Dietary additives should be provided in abundance and follow the guidance given earlier in the manuscript on adequate nutrition for YML, considering appropriate macronutrient profiles; this will help address malnutrition that results from feeding polymers to YML.

Future concern 2. Feeding of mycotoxin-contaminated grains to YML, resulting in reduced health

Agricultural products (particularly cereals) may be contaminated with mycotoxins produced by fungi/molds; high mycotoxin levels may cause grain to be destroyed. The IAFF industry is interested in the reclamation of mycotoxin-contaminated grains as feed for YML. Mycotoxin bioaccumulation, and/or the transformation of toxins into more dangerous compounds, can occur in YML (often at low levels; Niermans *et al.*, 2019; Ochoa-Sanabria *et al.*, 2022), representing a potential food/feed safety concern when rearing YML on contaminated substrates (Bordiean *et al.*, 2020; but see Duhra *et al.*, 2021; van Broekhoven *et al.*, 2017). Mycotoxins may have lethal or sublethal adverse effects on YM depending on toxin, toxin concentration, rearing conditions, species, exposure time, developmental stage, or the combination of mycotoxins produced by a fungal species (reviewed in detail in the text of Supplementary File S1 and see Tables 3 and 4).

Recommendation (domain 3: physical health): avoid feeding mycotoxin-contaminated grains to YML unless the mycotoxins present have been identified, their synergistic effects are known, and they have been diluted to levels known to be safe for YM consumption. Mycotoxin levels in naturally-occurring contaminated grains can be much higher than those tested in lab trials. For example, tests of the *Fusarium* spp. Mycotoxins, deoxynivalenol (DON) and zearalenone (ZEN), at concentrations up to 12,000 ug/kg yielded no strong negative effects (Tables 3 and 4). However, naturally infecting

feed with *Fusarium culmorum* produced DON levels of >10,000 ug/kg, and ZEN levels of 210,000 ug/kg; this resulted in reduced survival of last-instar YML (Guo *et al.*, 2014).

Several mycotoxins have been tested at higher concentrations in other species of Coleoptera. Dose-dependent mortality reached 100% for some species/toxin/concentration combinations (reviewed in Niermans *et al.*, 2021; Figure 3), suggesting that the high concentrations of toxins found in naturally-contaminated substrates may be more toxic to YML than the studies performed using lower levels currently demonstrate.

Further, many studies looked only at the effect of a single mycotoxin (Table 3), however naturally-occurring fungal contamination typically produces more than one mycotoxin (Table 4), which may synergistically increase toxicity (Canteri de Souza *et al.*, 2018). Studies that look at the effects of mycotoxin mixtures or naturally contaminated grain sources may be a more accurate representation of the welfare impacts of mycotoxin-contaminated substrates on YML.

If mycotoxin-contaminated substrates cannot be avoided, tests should be performed that ascertain which fungal species and mycotoxins are present, and if they have any known synergistic effects on YML. Producers should then appropriately dilute mycotoxin concentrations to protect welfare, productivity, and product safety.

Recommendation (domain 3: physical health): avoid feeding early-instar YML mycotoxin-contaminated grains

Many studies looking at the effects of mycotoxins on YML are conducted on later-instar larvae (Tables 3 and 4). However, early-instar larvae are expected to be more susceptible to mycotoxins (Abado-Becognee *et al.*, 1998); therefore, these studies do not necessarily demonstrate these feeds are safe for early-instar larvae.

Recommendation (domain 1: nutrition; domain 3: physical health; domain 4: behavioral restriction; domain 5: mental): do not feed YML mycotoxin-contaminated substrates that they behaviorally avoid, such as ergot-alkaloids or certain enniatins

Behavioral avoidance of contaminated substrates has been recorded (Table 4): YML reduced their intake of EA-contaminated feed over 21 days, despite no apparent preference/avoidance of contaminated feed after 20 minutes of exposure (Ochoa-Sanabria *et al.*, 2022). Avoidance behavior strongly correlated with toxin-induced mortality (Guo *et al.*, 2014). Behavioral avoid-

TABLE 3 Effects of different mycotoxins in artificially contaminated diets on YML welfare

Toxin	Concentrations in diet	Larval age	Effects of mycotoxin	Reference
AFB1	13-415 µg/kg	First instar; reared through pupation	Survival unaffected; slight increase in body mass when fed 204 µg/kg (effect not concentration-dependent), otherwise body mass unaffected	Bosch <i>et al.</i> , 2017
AFB1	100 µg/kg	3rd through 8th instars	Survival unaffected; weight gain unaffected for 30 days	Zhao <i>et al.</i> , 2022
DAS	0-128 ppm	~10 mg in weight; reared for 4 weeks	Survival decreases up to 20%, but not strongly dose-dependent; reduction in growth rate, food utilization	Davis and Schieffer, 1983
DON	8,000 µg/kg	5 weeks; reared 2 weeks	Survival unaffected; body mass unaffected	van Broekhoven <i>et al.</i> , 2017
DON	4,900-25,000 µg/kg	2 months old; reared through pupation	Survival unaffected; dose-dependent larval body mass and protein reduction; increased antioxidant enzyme activities; reduced locomotion	Janković-Tomanic <i>et al.</i> , 2019
Fumonisin B1	50,000-450,000 µg/kg	15 weeks old (injection study); 7 weeks old (feeding study), reared through pupation	Survival unaffected in all conditions. When injected, single exposure: acute reduction in metabolic rate, followed by recovery; When fed continuously: decline in growth at 450,000 µg/kg, increased larval instar duration, reduction in diet consumption, reduction in metabolic rate (all around 28 days of feeding)	Abado-Becognee <i>et al.</i> , 1998
T-2 toxin	0-128 ppm	~10 mg in weight; reared for 4 weeks	Survival unaffected; reduction in growth rate, protein utilization, feeding behaviors	Davis and Schieffer, 1982
T-2 toxin and HT-2 toxin	54 and 52 µg/kg	42 days old; reared 4 weeks	Potential increase in survival on high T2/HT-2 diet**;	Piacenza <i>et al.</i> , 2020
	140 and 120 µg/kg		potential increase in body mass on T2/HT-2 diets	
ZEN*	427.0-2,283 µg/kg	42 days old; reared 4 or 8 weeks	Survival unaffected; change in body mass unaffected	Niermans <i>et al.</i> , 2019

* Also contained DON: (control: 572 µg/kg) ranging from 568.4-2,101 µg/kg; other mycotoxins may be present and not analyzed.

** Statistics not provided for comparing artificial-control to artificial-contaminated diets.

TABLE 4 Effects of different mycotoxins in naturally contaminated diets on YML welfare

Fungal species	Main toxins*	Concentration in diet	Larval age	Effects of mycotoxin	Reference
Not listed	BEA	13 µg/kg	5 weeks; reared 2 weeks	Survival unaffected; increased body mass	van Broekhoven <i>et al.</i> , 2017
	DON**	4,900 µg/kg			
	Eas	21 µg/kg			
	ENAI	45 µg/kg			
	ENB	490 µg/kg			
	ENBI	170 µg/kg			
	NIV	270 µg/kg			
Not listed	ZEN	73 µg/kg	42 days old; reared 4 or 8 weeks	Survival unaffected; increased growth rate compared to control and artificial diets	Niermans <i>et al.</i> , 2019
	DON	2,854 or 4,588 µg/kg			
	ZEN	602.3 or 919.3 µg/kg			
Not listed	Eas	(control: 63 µg/kg)	7th-9th instar; reared 21 days	Survival unaffected; growth rate unaffected; reduced feed intake; no preference/avoidance	Ochoa-Sanabria <i>et al.</i> , 2022
Not listed; 'ergot'	Not assessed	3,863-15,316 µg/kg	~10 mg larvae; reared for four weeks	20 minutes after initial exposure	Davis <i>et al.</i> , 1975
<i>F. culmorum</i>	DON ZEN	10,240 µg/kg	Last instar, reared 15 days	Survival unaffected; growth rate reduced in 20% compared to control	Guo <i>et al.</i> , 2014
<i>F. avenaceum</i>	ENA	>210,000 µg/kg	Last instar, reared 15 days	Reduced survival; increased body mass; behavioral preference	Guo <i>et al.</i> , 2014
	ENAI	14,260 µg/kg		Reduced survival; reduced body mass; behavioral avoidance	
	ENB	60,510 µg/kg			
	ENBI	>90,000 µg/kg			
<i>F. poae</i>	BEA	>90,000 µg/kg	Last instar, reared 15 days	Survival unaffected; increased body mass; strong behavioral preference	Guo <i>et al.</i> , 2014
	ENB	30,100 µg/kg			
	ENBI	320 µg/kg			
<i>F. proliferatum</i>	BEA	30 µg/kg	Last instar, reared 15 days	Survival unaffected; increased body mass; strong behavioral preference	Guo <i>et al.</i> , 2014
	ENA	36,360 µg/kg			
	ENB	30 µg/kg			
	ENBI	27,090 µg/kg			
	FBI	2,890 µg/kg			

TABLE 4 (Continued)

Fungal species	Main toxins*	Concentration in diet	Larval age	Effects of mycotoxin	Reference
<i>F. graminearum</i>	DON†	(control: 200 µg/kg) 2,000–12,000 µg/kg	Whole life cycle	Survival to pupation unaffected, pupal survival reduced; body mass of larvae unaffected; reduction in adult body weight of the second generation; no preference/avoidance	Ochoa-Sanabria et al., 2019
<i>F. sporotrichioides</i>	T-2 toxin HT-2 toxin	11 and 89 µg/kg 26 and 262 µg/kg	42 days old; reared 4 weeks	Survival unaffected; increased body mass	Piacenza et al., 2020

* Abbreviations: BEA = Beauvericin, DON = Deoxynivalenol, EA = Ergot Alkaloids (all types), EN = Enniatins, FBI = Fumonisin BI, NIV = Nivalenol, ZEA = Zearealenone. In all cases, other toxic products may be present and were simply not analyzed.

** Modified DON present: Deoxynivalenol-3-glucoside – 300 µg/kg; 15-acetyldeoxynivalenol – 86 µg/kg

† Modified DON present: 3-acetyldeoxynivalenol – <LOD, 63, 52, 205 µg/kg per diet.

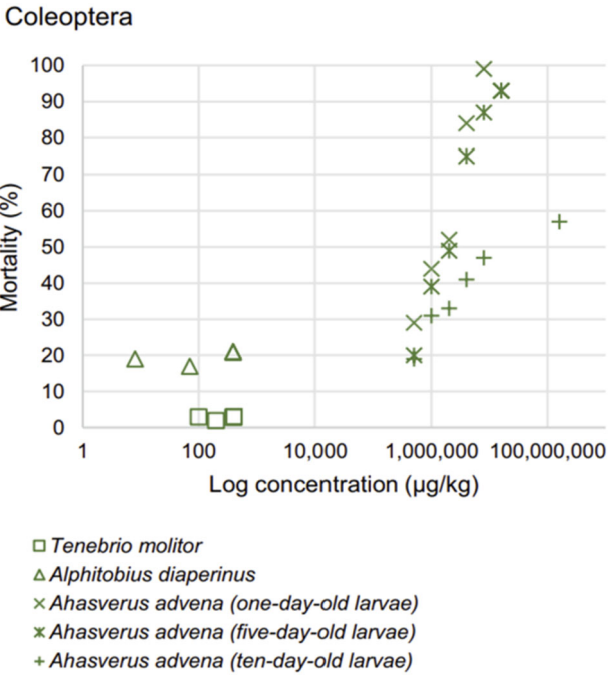


FIGURE 3 Mortality (%) caused by exposure to different doses of AFB1 (log scale) for Coleoptera. FROM NIERMANS ET AL. 2021 (CC-BY-4.0)

ance of fungal contaminants may be the mechanism driving sublethal declines in body size, suggesting malnutrition may be another welfare concern (Howard, 1955). Reduced physical health, potential malnutrition, and reduced behavioral expression (indicated in a reduced desire to feed) may also result in effects on the mental domain, such as hunger.

Future concern 3. Genetic modification or selective breeding that results in reduced health or behavioral restriction

Genetic modifications are being explored for farmed YM, including the use of CRISPR-Cas9 (Chu, 2019) and RNA interference (RNAi; Oppert et al., 2019), which may involve rendering specific (presumably undesirable) genes non-functional or introducing novel (presumably desirable) genes from hetero- or conspecifics (reviewed in Errikson and Picard, 2021). These manipulations were made possible in part by the publication of the YM genome (Eriksson et al., 2020). The most likely genetic modifications explored for YM are those that would increase economic productivity, for example by: increasing growth rates or feed utilization, speeding up reproduction or development, increasing survival or fecundity, or generating better disease resistance. Undesirable characteristics, such as traits that increase allergenicity (Ribiero et al., 2021) may also be targeted for removal.

Selective breeding (i.e. artificial selection) to enhance production is perhaps of more immediate interest to the industry (Tomberlin *et al.*, 2022). Selective breeding has historical precedent and can be done particularly efficiently in r-strategist species (life history characterized by high fecundity, small body size, short generation time, etc.) like the YM (Eriksson and Picard, 2021). Strains within rearing facilities are already differentiating phenotypically (presumably, due to both natural and artificial selection, as well as neutral evolutionary processes), with documented variation in final larval body mass, development time, survival, and feed utilization (Adamaki-Sotiraki *et al.*, 2022; Rumbos *et al.*, 2021).

Recommendation (domain 3: physical health; domain 4: behavior): any novel strains should be studied for the progressive impact of modifications on welfare across the lifespan, prior to large-scale adoption by the industry. Strains with worse welfare should be avoided. Selective breeding has been responsible for many welfare problems in vertebrate livestock production (Paxton *et al.*, 2013) and genetic management always carries some welfare risks (Ormandy *et al.*, 2011). Researchers and producers should take care to evaluate the welfare impacts of any genetic management initiatives in their facilities.

Complex effects can result from genetic modification; consider the recent use of RNAi to knockdown YM *vermilion*, producing white eyes in YM. *Vermilion*-knockdown in YM affected the expression of 109 other genes, including a reduction of Aspartate 1-decarboxylase (ADC). Decreased ADC expression in flour beetles resulted in behavioral defects, including a reduction in movement speed and distance and an increased amount of time spent immobile (Perkin *et al.*, 2017). Genetic modifications may thus have unintended physiological or behavioral consequences that manifest as animals age (Dennis, 2002) and harm the animals' welfare.

Two artificial selection experiments (7-8 years long) selected for larger late-stage YM larval body mass, reduced development times, greater conversion efficiency of ingested food, and higher egg production (LeClercq, 1963; Morales-Ramos *et al.*, 2019). However, the aforementioned benefits were counteracted by reduced survival in the selected strain, resulting in similar biomass production overall. Larval survival was 10.5% in the selected strain and 24.2% in the ancestral strain by 60-69 days. Approximately 9% of the artificially selected larvae died *after* six weeks of development (Morales-Ramos *et al.*, 2019), suggesting the

negative physiological states that are eventually lethal in this strain may be experienced for a prolonged duration prior to mortality. It is thus possible that trait tradeoffs (in this case, between larval survival and mass) may provide no benefit to economic productivity while harming welfare.

Prior to deploying any new strains at scale, researchers should confirm there are no adverse physiological or behavioral challenges as a result of the genetic manipulation. All life stages should be studied as negative welfare consequences may only manifest during some stages of development.

Recommendation (domain 3: physical health; domain 4: behavior): explore the use of genetic tools to enhance farmed YM welfare

There is also an opportunity for genetic modification or selective breeding to enhance farmed YM welfare. For example, selection for 'black' lines of YM adults correlated with improved disease resistance (Armitage and Siva-Jothy, 2005; Armitage *et al.*, 2003). Selective breeding fixed the 'black' cuticular color in YML over three generations with only a slight reduction in late-stage larval mass (Song *et al.*, 2022; and see Huang *et al.*, 2011). Selective breeding could thus improve insects' resistance to immune challenges faced in high-density farmed conditions, with concomitant benefits for welfare. Other researchers have argued that there may be a technological solution to the problem of farmed animal suffering, where livestock should be genetically engineered with a reduced or, if possible, eliminated capacity to suffer (Shriver, 2009; though see other arguments against genetic modification of livestock animals, such as commodification and unnaturalness, in Bovenkerk, 2020). Significant exploration of the possible positive welfare impacts of genetic management for YM (for instance, reducing cannibalistic behavior or enhancing disease resistance) is therefore needed.

Future concern 4. The use of diatomaceous Earth when rearing YM adults

Diatomaceous earth may be used to inhibit egg cannibalism of adult beetles (e.g. Shostak, 2014). However, this may result in negative welfare consequences if (1) abrasions are severe enough to reach nociceptive sensory cells underlying the cuticle, (2) dehydration occurs, as the cuticular hydrocarbons that reduce desiccation stress are eroded; or (3) levels are high enough to induce mortality.

Recommendation (domain 3: physical health; domain 5: mental): avoid the use of diatomaceous Earth when rearing YM adults

Other methods should be explored to reduce egg cannibalism by adults, as diatomaceous earth may result in reduced health, mortality, and possibly pain or discomfort.

7 Urgent future research directions in yellow mealworm welfare

A recent review on the evidence for insect pain/sentience demonstrated that there were particularly significant evidence gaps for the Coleoptera at both the larval and adult life-stage (Gibbons *et al.*, 2022). To better understand YM welfare across development, more work on sentience-relevant neurobiological features and behavioral abilities (e.g. according to the Crump *et al.*, 2022 framework) of YMs is needed. Additionally, there is presently no empirical work that has validated behavioral or physiological indicators of insect welfare for YM or other species. These data are also needed so that welfare in intensive production facilities can be assessed.

Standard methods for rearing YM likely cause some negative welfare impacts. However, there are many cases where we reviewed the behavioral and / or physiological data and were not able to determine yet whether conditions negatively affect YM welfare. Further, almost no information was found about how to promote positive welfare for YM in the course of our review. The following empirical questions emerged out of the comprehensive review of YM biology and welfare found in Supplementary File S1. The following questions are most essential for improving our understanding of YM welfare in farmed conditions (in no particular order):

- *Waste feed safety*: One of the essential promises of the IAFF industry is the ability to create more sustainable, circular agricultural systems through the transformation of waste feeds into usable protein (in the form of insects; van Huis and Gasco, 2023). However, as discussed in this review and Supplementary File S1, many of these waste feeds contain toxins, pesticides, heavy metals, or pathogenic contaminants that are, or may be, undesirable for welfare. A top priority for insect welfare researchers should be determining what pre-processing is required to make any feeds that have the potential to be nutritious for insects also safe for their health. This may include heat-treatments for reducing disease-causing agents fol-

lowed by inoculation with probiotics, mixing waste and non-waste streams to provide the right macronutrients or reduce toxin/pesticide/heavy metal levels, etc.

- *Disease/parasitic agents*: It is essential to identify, prevent, and proactively manage potential disease/parasitic agents, including ones present at sublethal levels, for both animal welfare and economic productivity (Tomberlin *et al.*, 2022). Researchers should also study ways of promoting animal agency and positive welfare regarding disease – through abiotic gradients (e.g. temperature gradients that allow for behavioral fever in insects), logistical choices (low-density rearing environments), and techniques like transgenerational immune priming.
- *Cannibalism*: Cannibalism is a serious problem in the industry that results in injury, disease, and mortality. It is crucial to have studies that demonstrate how volumetric and aerial stocking density, substrate depth, larval rearing practices, selective breeding, hydration/RH, and nutrition contribute to YML cannibalism.
- *Stocking density*: Stocking conditions may contribute to overheating, disease, behavioral restriction, cannibalism, malnutrition through resource competition, and more. Research should carefully tease apart the effects of variables like aerial vs volumetric density, substrate depth, and the effects of physical contact, heat, consumption of frass, and resource competition, to determine the optimal stocking conditions for YML.
- *Feed efficiency*: Because YML live *within* their feeding substrate, feed efficiency concerns are frequently an argument for higher stocking densities which, in turn, raise welfare concerns. Research is needed to determine if: (1) incorporating inedible substances into the substrate could serve to reduce stocking density while avoiding feed waste; (2) changing crate designs could improve larval dispersal throughout the provided substrate, reducing localized density while again managing feed waste.
- *Larval diets*: Several outstanding questions remain for when considering larval nutrition. Further work is needed on the welfare benefits of fats and proteins in the diet, especially whether these components help reduce larval cannibalism. Further, no research has focused on the welfare effects of 'batch' or more 'progressive' feeding methods for YML or whether overfeeding has negative welfare effects (as has been observed in other larvae).

- *Mycotoxins*: It would be valuable to have studies that assess the welfare impacts of natural sources of mycotoxin-contaminated grains on YML, particularly at early instars that may be more sensitive to the toxins. It is especially important to further test high concentrations and grains with multiple toxins present.
- *Adult YM*: Adult YM welfare is particularly understudied. Researchers should prioritize inquiry into optimal diets (including the possibility of adult protein limitation under current conditions; see Supplementary File S1), stocking densities, and abiotic conditions, as well as threat vectors such as diseases and parasites.
- *Slaughter and depopulation*: Very little is known about the humaneness (intensity of pain $\times$ time to death, where less is better) of YM slaughter and depopulation methods under different standard operating procedures. For instance, freezing in air is used to cull adult YM and slaughter YML. However, the welfare impacts of freezing in air at industry scales are largely unexplored and the humaneness of the practice is currently debated.

Further, research that answers these less-urgent but still important questions would also improve our understanding of farmed YM welfare:

- *Mites*: Mites are highly prevalent in most production facilities; their welfare impacts are unknown. Research on the welfare effects and possible prevention of the most common mites would be beneficial.
- *Genetic modification and artificial selection*: The progressive physiological and behavioral effects of genetic modifications/artificial selection across the lifespan should be tested prior to the adoption of new lines by the industry. The industry may also choose to research genetic management programs that could improve insect welfare, such as lines selected to reduce cannibalistic behavior.
- *Probiotics*: Some probiotics may support positive welfare through improved immune health and nutrient utilization if the feed is inoculated prior to YML consumption; further, some fungal secondary metabolites may be beneficial. Probiotic research may start with *P. pentosaceus*, *B. subtilis*, *B. toyonensis*, and *E. faecalis*, which have already demonstrated potentially positive effects as food additives, while *F. poae*, *F. proliferatum*, and *F. sporotrichioides* may be worth investigating on the fungal side (Table 4).
- *Giant YML*: It is currently unknown whether the practice of rearing giant YML through supplemental juvenile hormone in feed restricts behaviors or harms physical health (see review in Supplementary File S1). Researching the behaviors and physiology of giant and normal YML, particularly in relation to traits associated with increased fat body size in other insects (e.g. heart dysfunction, hyperglycemia, reduced activity levels, and insulin resistance), will be essential to determining whether this practice harms YM welfare.
- *Overheating*: It may be possible to avoid localized larval overheating by using alternative crate materials and substrates, reducing YML densities, and employing monitoring devices – which would improve both welfare and productivity.
- *Light*: Research is needed on the welfare effects of photoperiod, and the type of light provided, for rearing YM at all life stages (see Supplementary File S1).
- *Handling*: Handling-associated stress (due to light, fasting/washing, vibrations, etc.) should be assessed for potential negative welfare effects. Additionally, producers should test if particular practices can be less frequent, shortened, or discontinued without affecting product quality/safety (e.g. fasting). Acclimation to handling may also occur in adult YM, resulting in reduced stress over the course of the lifespan; acclimation programs could be piloted for adult beetles.
- *Anesthetics*: Appropriate anesthetics, and industry-scale delivery methods, should also be researched further, including their effects on product quality and safety.
- *Prevalence of welfare concerns*: No data are available on the prevalence of most welfare concerns in industry conditions. Likewise, no data are available on the practices producers employ to maintain high welfare conditions for their animals. Research that documents prevalence and assesses how changes in rearing practices affect welfare will be essential for guiding high welfare standards and increasing economic productivity.

8 Conclusion

YM have been reared commercially in the US since the 1950s, though not at significant scale. In recent years, the IAFF industry has increased production capacity at unprecedented rates. This growth in intensive mini-livestock production has prompted a corresponding growth in concern for the welfare of insects as individuals (Barrett *et al.*, 2022; Bear, 2019, 2021; Boppré and Vane-Wright, 2019; de Goede *et al.*, 2013; Delvendahl *et al.*, 2022; Erens *et al.*, 2012; Fukada *et al.*, 2023; Gjerris *et*

al., 2016; IPIFF, 2019; Kortsmit *et al.*, 2022; Lambert *et al.*, 2021; van Huis, 2021).

Although this comprehensive review aims to help YM producers engage in high-welfare rearing and slaughter practices based on current understanding of the species, insect welfare, and industry practice, there are clearly many important gaps in our knowledge. We continue to echo calls by producers (e.g. IPIFF, 2019) for significant further research on insect welfare that is relevant to the IAFF context. Further, the rapid growth of industry technologies and scale are likely to produce new, unanticipated welfare challenges (Barrett and Fischer, 2023). As our knowledge of insect welfare grows and practices change, this information within review will require continual updating to maintain high quality, empirically-based welfare practices for Yms across life stages. High-impact research on insect welfare requires several kinds of resources to move the field forward: (1) funding, (2) industry-scale research facilities and (3) transparent knowledge-sharing and collaboration between producers, entomologists, and welfare scientists to address emerging challenges (Barrett and Fischer, 2023; Barrett and Adcock, 2023).

The IAFF industry is already value-driven: its members are committed to environmental sustainability, economic development, food security, democratizing food production, resource stewardship, and many other values. Including insect welfare on this list aligns with the IAFF industry's self-understanding as an ethically responsible enterprise. By collaborating with a wide array of researchers and stakeholders, the industry can preemptively address the familiar welfare issues that have arisen in many concentrated *vertebrate* animal feeding operations. Instead of responding to unanticipated challenges after major investments in infrastructure, when change is expensive, the industry can design systems with welfare in mind, thereby providing insects with optimal conditions throughout the lifecycle.

Supplementary material

Supplementary material is available online at: <https://doi.org/10.6084/m9.figshare.24647805>

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Conflict of interest

Meghan Barrett, Bob Fischer, Alexandra K. Schnell, and R. Keating Godfrey report financial support was provided by Rethink Priorities. Bob Fischer reports a relationship with Rethink Priorities that includes: employment. R. Keating Godfrey, Alexandra K. Schnell, and Meghan Barrett report a relationship with Rethink Priorities that includes: consulting or advisory.

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